

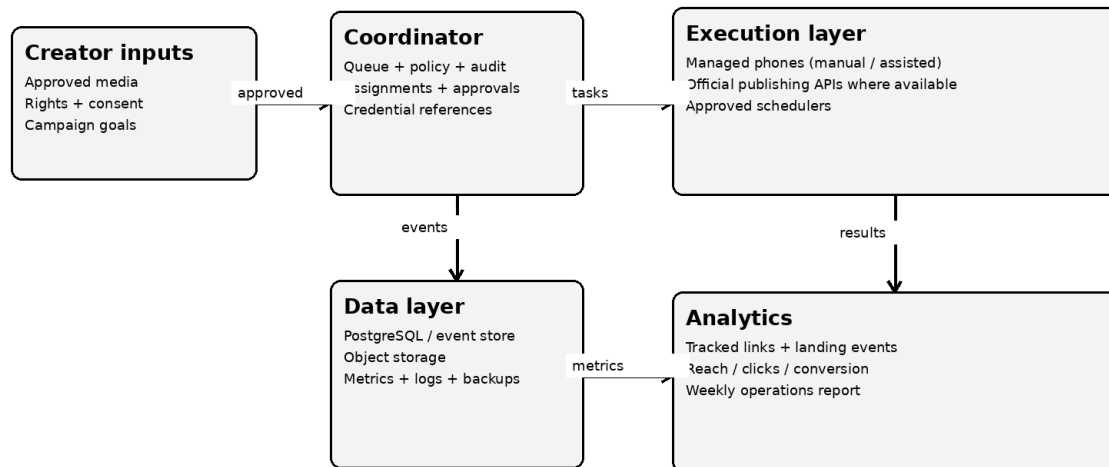
OFM PHONE FARM ENGINEERING GUIDE

Physical infrastructure, managed-device fleets, networking, software control planes, publishing workflows, analytics, security, reliability and operational governance

Research edition - 26 August 2026

Reference Architecture

A compliant, observable creator-traffic operations platform



Reference architecture used throughout this guide.

Scope: legitimate creator-traffic operations using organization-owned devices, official management tooling, authorized accounts, approved publishing paths and auditable employee workflows.

Reader notice and safety boundary

This manual explains phone-farm engineering as a fleet-operations problem. It covers hardware, power, thermal design, USB, Wi-Fi, cellular concepts, NAT, CGNAT, DNS, VPNs, forward proxies, MDM, Android Enterprise, ADB, device/coordinator communications, software architecture, content operations, analytics, security, staffing, incident response and scaling. It is written for an OFM or creator-management environment, but most engineering material also applies to QA labs, retail kiosks, field-device fleets and social-media production teams.

What is intentionally not operationalized

The guide does not provide recipes for defeating platform enforcement, concealing a device or account fingerprint, falsifying geographic location, rotating identities to evade bans, automating account creation at scale, manufacturing fake engagement, bypassing rate limits, defeating CAPTCHAs, or scripting third-party social UIs to imitate humans where the platform prohibits automation. Those techniques create platform, fraud, security and contractual risk. Proxy/VPN and device-identity topics are therefore explained for networking, privacy, segmentation, testing and observability - not evasion.

Some OFM-specific claims circulating in operator communities are anecdotal. They are labeled as such in this guide. Technical claims about fleet management, device provisioning, power, temperature, publishing APIs and platform rules are anchored primarily to first-party documentation listed in the References section. Platform rules change; re-check them before deployment.

How to use this guide

If you are...	Start with	Then read
Owner / agency principal	Ch. 1-5, 55-61	Costs, risks, operating model, roadmap
Infrastructure engineer	Ch. 6-23	Power, rack, Wi-Fi, routing, VPN/proxy/DNS
Mobile/fleet engineer	Ch. 24-31	Apple MDM, Android Enterprise, ADB, enrollment
Backend engineer	Ch. 32-41	Coordinator, queues, APIs, database, telemetry
Operations manager	Ch. 42-54	Content pipeline, employees, analytics, SOPs
Security/compliance	Ch. 4, 52-59	Access, logs, incidents, privacy, offboarding
Builder starting a pilot	Ch. 60-65 + App. A-I	BOM, acceptance tests, build sequence

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PART I - PURPOSE, INDUSTRY MODEL AND BOUNDARIES

A phone farm is only useful when the business process around it is explicit. This part defines the OFM use case, separates legitimate fleet operations from manipulation, and establishes the architecture used throughout the manual.

1. Phone farms in OFM

In an OFM context, a phone farm is a managed collection of physical mobile devices used by an agency or creator team to operate authorized social accounts, prepare and publish content, review notifications, collect performance signals and support traffic acquisition. The physical phones are execution endpoints; the real system is the combination of devices, people, network, content rights, credentials, workflow software and analytics.

Industry usage and terminology

Public operator communities use “phone farm” loosely. It can mean ten old iPhones on a desk, a rack of Android phones, a motherboard rack without screens, a set of remotely controlled devices, or a larger operation with coordinators and staff. These reports are anecdotal and should not be treated as evidence that a particular platform will tolerate the associated behavior.

The legitimate engineering interpretation is closer to a corporate mobile fleet or device lab: each device is inventoried, assigned, secured, connected to controlled networks, monitored and maintained. The difference is that the workload is creator-content operations rather than QA testing or retail kiosk use.

- Define the owner of every device and social account.
- Treat creator authorization and content rights as first-class records.
- Separate device administration from social publishing.
- Measure output with business metrics rather than “number of phones.”

What the farm is not

A physical phone is not an invisibility device. Platforms can evaluate account behavior, content similarity, network characteristics, login history and many other signals. A real handset may make the workflow more operationally convenient, but it does not convert prohibited behavior into permitted behavior.

The architecture should therefore optimize for repeatability, security, operator throughput, reliability and attribution. It should not optimize for hiding the existence of the fleet.

- Do not buy hardware before mapping the traffic funnel.
- Do not assume one physical device must equal one account. The correct mapping depends on authorization, platform rules, staffing and operational risk.

Engineering checklist

- Write down the business goal for each platform.
- List authorized creator accounts and account owners.
- Define which actions are manual, API-driven or scheduler-driven.
- Create a platform-rule register with review dates.

2. Traffic funnel and business model

The phone-farm layer sits in the middle of a larger creator acquisition funnel. Engineering decisions should be evaluated by their effect on measurable funnel stages rather than raw posting volume.

Reference funnel

A typical funnel is: approved creative asset -> platform post -> profile visit -> tracked link or landing page -> creator page -> subscription or other paid action. Depending on the business, there may be additional layers such as a link hub, geofenced landing page, email capture, or CRM attribution. Every added layer creates potential conversion loss and additional privacy/security obligations.

The control plane should attach campaign, creator, platform, account, device, operator and asset identifiers to each publishing task. The analytics plane should connect those identifiers to post-level reach where available and to downstream click/conversion events where lawful and technically possible.

- Primary operational metrics: task completion rate, post success rate, account availability, content approval latency.
- Primary marketing metrics: reach, profile visits, click-through rate, landing conversion, subscriber conversion, revenue per thousand impressions where calculable.
- Primary reliability metrics: device heartbeat, battery/thermal status, network packet loss, API error rate, queue age.

Unit economics

A useful model treats each account or campaign as a cost center. Hardware amortization, software, connectivity, employee time, content production and creator revenue share should be visible. "More phones" only creates value if incremental gross contribution exceeds incremental infrastructure and labor cost.

Attribution is imperfect because not every platform exposes the same analytics and some downstream conversions cannot be deterministically tied to a post. The system should distinguish observed events from modeled or inferred attribution.

Layer	Example measure	Engineering owner
Creative	Approved assets per week; QC rejection rate	Content ops
Publishing	Task success; queue age; retries	Fleet/backend
Platform	Reach; profile actions	Growth ops
Landing	Unique clicks; conversion	Web/backend
Revenue	Paid conversion; ARPU/LTV	Finance/analytics

3. What physical phones solve - and what they do not

Physical devices are valuable when the workflow actually requires the native mobile app, hardware-bound capabilities, mobile camera/media access, or a standardized operator interface. They are less valuable when an official API or web console already provides the required function.

Control Plane vs Data Plane

Do not mix fleet administration with creator content and social traffic

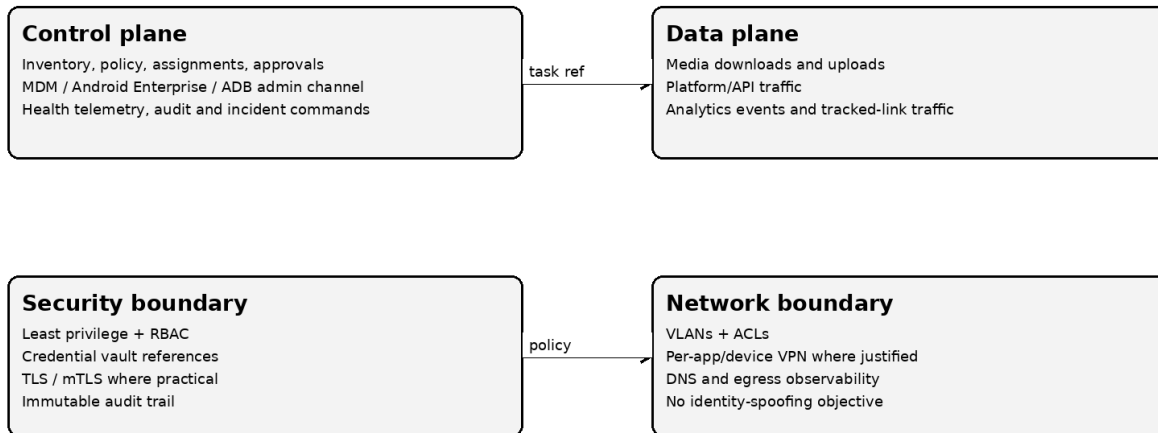


Figure 3-1. Control-plane and data-plane separation.

Problems physical devices solve

Phones provide the authentic application runtime, camera roll, push notifications, biometric-secured credential storage, mobile-only creation tools and native moderation/status screens. Organization-owned devices can also be physically segmented from employees' personal phones.

For Android, corporate devices can be fully managed or dedicated under Android Enterprise. For Apple fleets, organization-owned supervised devices can receive stronger MDM controls than unsupervised personal devices. [S1][S5]

- Native-app-only features.
- Consistent operator work surface.
- Separation between personal and corporate credentials.
- Physical inventory that can be wiped, reassigned and retired.

Problems they do not solve

Phones do not remove platform terms, content restrictions, API rules or behavioral enforcement. They also do not automatically solve account ownership, credential leakage, employee mistakes, attribution or creator-consent management.

A rack can even make risk worse if the same content is repetitively distributed, credentials are broadly shared, devices are not patched, or network architecture creates one large blast radius.

- Build authorization and access control before scaling hardware.
- Use official publishing APIs where available and suitable.
- Keep humans in approval loops for ambiguous or high-risk actions.

4. Platform-policy and authorization boundary

The most important non-hardware requirement is an explicit rule boundary. Different platforms permit multiple accounts, third-party management, APIs and scheduling to different degrees, but commonly prohibit spam, fake engagement, coordinated inauthentic activity, automated bulk account operation, or circumvention of enforcement. These rules can change.

Policy update rule

Treat all platform-specific operating rules in this manual as a dated snapshot, not a permanent guarantee. Re-check first-party policies and API terms before deploying a workflow change.

Current examples from first-party rules

TikTok's integrity rules distinguish authentic multi-account use from deceptive or spam behavior and prohibit automation used to operate accounts in bulk or manipulate platform systems. TikTok also provides a formal Content Posting API for authorized applications, with scopes and audit requirements. [S22][S23]

Reddit's rules require authentic participation and prohibit spam and manipulation. Snapchat explicitly prohibits spam and artificially inflated engagement. X permits legitimate third-party account management but prohibits coordinated inauthentic activity, duplicate amplification and ban-evasion behavior. [S24][S26][S27]

Authorization records

For every creator and account, retain a signed or otherwise auditable authorization record stating who owns the account, what the agency may do, which employees or systems may access it, what content is approved, and what happens on termination. The agency should not rely on informal chat messages as its only control evidence.

Keep platform credentials separate from legal identity documents and sensitive creator files. The system should store credential references to a vault rather than passwords in application tables or spreadsheets.

- Creator ID and contract/authorization reference.
- Account owner and recovery email/phone owner.
- Permitted actions and publishing platforms.
- Content usage scope and expiration/revocation process.
- Offboarding/wipe/recovery procedure.

5. Architecture decision: API-first, device-first or hybrid

The correct architecture depends on which functions are officially exposed by platform APIs and which require a person in a native application. A hybrid design is usually the most maintainable: APIs or approved schedulers handle supported publishing and analytics; physical phones handle native-only tasks and human review.

Reference Architecture

A compliant, observable creator-traffic operations platform

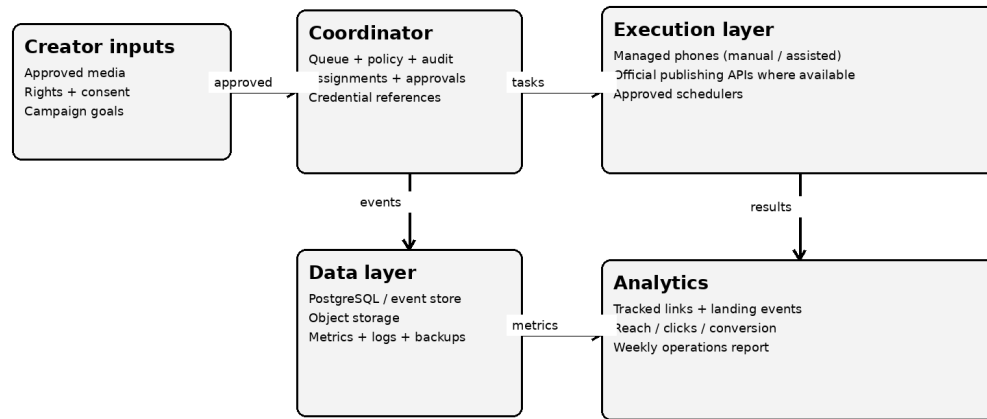


Figure 5-1. Hybrid reference architecture.

API-first

Use an API-first workflow when the platform provides a supported publishing or analytics endpoint and the agency can meet app-registration, scope and audit requirements. API-first systems are easier to log, retry, rate-limit, test and attribute. TikTok's Content Posting API is one example of an explicit, authorized posting path. [S22]

- Advantages: deterministic requests, auditability, fewer physical touch points, easier scaling.
- Costs: developer integration, platform app review, scope limitations, changing APIs.

Device-first

Device-first means a human operator performs most social actions on managed phones. This is slower to scale but can be appropriate for native-only workflows. Fleet software should coordinate assignments and record outcomes without pretending to be the human.

If the platform does not permit automated UI operation, do not turn accessibility tools, ADB input commands or computer vision into a covert substitute for an official API.

- Advantages: access to native UX; human judgment.
- Costs: labor, training, ergonomic complexity, more devices, weaker deterministic automation.

Hybrid

Hybrid systems place a coordinator above both paths. A task can route to an official API executor, an approved scheduler, or a human/device queue depending on platform, account and action type. All paths write back a common result schema.

Dimension	API-first	Device-first	Hybrid
Scale	High where API permits	Labor-bound	High for supported actions
Auditability	Strong	Requires operator capture	Strong if unified
Native-only features	Limited	Strong	Strong
Platform dependence	API contracts	App UX changes	Both
Recommended role	Default where supported	Exception/native tasks	Overall architecture

PART II - PHYSICAL AND ELECTRICAL ENGINEERING

A rack of phones is a continuous electrical and thermal load, not a collection of casual chargers. This part treats the installation as small infrastructure: load planning, serviceability, airflow, cable discipline and lifecycle management.

6. Device selection

Device choice should be driven by supported OS life, battery condition, application compatibility, management support, reparability and total cost rather than the cheapest used handset price.

Selection criteria

Prefer models that remain inside the OS/security support window for the expected service life. Confirm the required social apps still support the OS version. For Android, verify Android Enterprise management capabilities and consider zero-touch eligible procurement for larger deployments. For Apple, organization-owned devices that can be enrolled and supervised through the intended deployment path are easier to administer consistently. [S1][S5]

Used devices can lower capital cost but introduce unknown battery health, replaced parts, activation locks, inconsistent storage and shorter remaining update life. A receiving inspection process is therefore mandatory.

- Supported OS/security updates for planned life.
- 64 GB or more storage if local media caching is heavy; size to actual workflow.
- Healthy battery and charging port.
- Stable Wi-Fi radios and compatible bands.
- Consistent model families to reduce spare-parts and cable variety.
- No activation lock, unknown MDM enrollment or carrier financing lock.

Screened vs headless hardware

Phone-motherboard racks and screenless devices appear in operator communities, but they add maintenance and compliance uncertainty because the hardware may be modified or difficult to recover. For a professional pilot, normal complete phones are easier to inventory, reset, update, visually inspect and hand to an operator.

Only consider custom hardware after the software architecture and workload are proven on standard devices.

Factor	Pilot preference	Scale preference
Model mix	1-2 models max	Standardize by cohort
OS age	Current or recent	Long remaining support
Battery	Known healthy	Replaceable service process
Procurement	Reputable used/new	Enterprise/reseller where possible
Management	Manual + MDM/Enterprise	Automated enrollment

7. Rack and mechanical design

The mechanical objective is not maximum density. It is stable airflow, easy identification, fast replacement, safe cabling and enough access that an operator can service one phone without disturbing ten others.

Mechanical Rack Concept

Serviceable layout: airflow, labeling, power distribution and cable relief

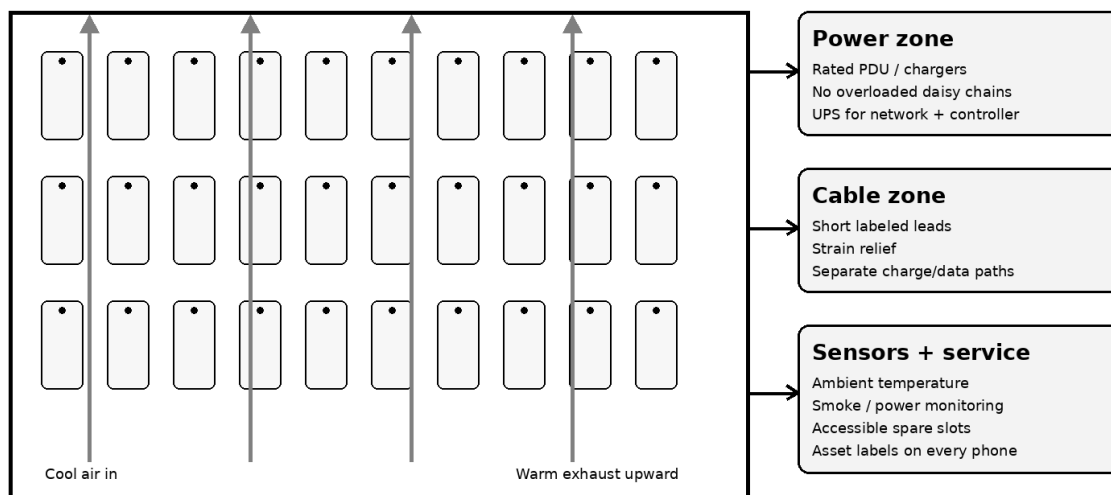


Figure 7-1. Example serviceable rack layout.

Slot geometry

Use vertical or slightly tilted slots so screens remain visible and warm air can rise between devices. Leave air gaps instead of clamping phones tightly together. The required spacing depends on device power, ambient temperature and fan airflow; validate it with temperature measurements rather than a fixed internet rule.

Every slot should have a stable slot ID. The physical label, inventory database and cable label should use the same identifier. Avoid relying on “third phone from the left.”

- Front access to power button and charging connector.
- No sharp edges or pressure on battery area.
- Cable bend radius and strain relief.
- Fan guards and removable dust filters where appropriate.
- A service aisle or pull-out tray for dense racks.

Material choices

Nonconductive trays simplify electrical isolation. Metal enclosures can provide stronger structure but require proper grounding and careful cable protection. 3D-printed fixtures should be made from suitable materials and kept away from hot power components; do not assume hobby plastics are fire-rated.

Keep chargers, AC distribution and high-current DC distribution in a power zone separate from phone slots. This improves cooling and makes fault isolation easier.

Engineering checklist

- Assign one immutable slot ID to every physical position.
- Provide a safe way to remove one device without pulling adjacent cables.
- Measure inlet and exhaust temperature under full load.
- Document rack weight, power-feed rating and emergency disconnect location.

8. Power architecture

Power should be designed from measured load. Charging power is dynamic: a phone may draw more while recovering from low battery, then taper as it approaches full charge, and thermal controls can reduce charging rate. USB Power Delivery negotiates power between compatible source and sink; USB PD 3.1 supports substantially higher power levels than phones normally require, so charger nameplate wattage is not the same as actual phone draw. [S10]

Illustrative Power Scaling

Planning example only - not a specification for any particular phone model

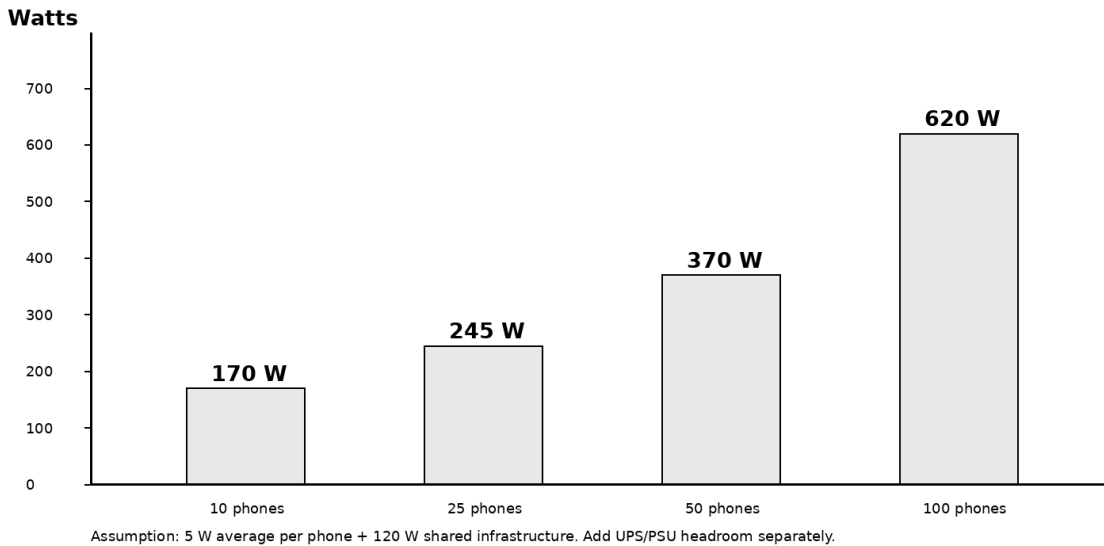


Figure 8-1. Illustrative power scaling model; use measured values for design.

Load model

For planning, define P_{devices} as the sum of measured average phone power, P_{peak} as the likely coincident peak, and P_{infra} as routers, switches, APs, controller PCs, fans and monitoring. Select supplies, branch circuits and UPS capacity with engineering headroom rather than operating continuously at nameplate limits.

Illustrative example only: if 25 phones average 5 W while connected, device load is about 125 W. Add the actual measured infrastructure load and then add headroom. This example is not a specification for any handset.

- Measure at the AC input with a power meter during representative use.
- Record both idle/maintenance load and recovery-from-low-battery load.
- Do not assume all multi-port chargers can deliver their advertised maximum to every port simultaneously.
- Use chargers/cables from reputable vendors with appropriate certifications for your region.

Power zoning

Split large racks into independently protected zones. A single failed adapter or tripped protection device should not remove the whole fleet. Network, control server and essential monitoring may deserve UPS runtime even if phone charging itself is not fully UPS-backed.

If chargers expose smart power allocation, test how port behavior changes when many devices are connected or disconnected at once.

Engineering checklist

- Create a port-to-slot map.
- Label every power supply with zone and rated output.
- Avoid improvised high-current wiring.
- Document shutdown/startup sequence after utility failure.

9. USB topology and cabling

USB can carry power, data or both. A rack that uses ADB over USB or Apple configuration tooling has different requirements from a charge-only rack. Treat the USB data topology as an engineered bus, not an unlimited fan-out.

Data vs charge

Charge-only cables reduce accidental data exposure but cannot support local USB administration. Data-capable cables are required for USB ADB and some provisioning workflows. Mark cable type visibly because identical connectors can have different capabilities.

Powered USB hubs should have their own adequate supplies. Avoid long chains of hubs; every tier adds connectors, failure points and potential bandwidth or enumeration problems. For large fleets, distribute devices across multiple host USB controllers and powered hubs rather than one massive tree.

- Use short, labeled cables.
- Maintain a spare cable bin by exact type.
- Log recurring disconnects by slot; intermittent cables cause disproportionate troubleshooting time.
- Use strain relief so the phone connector is not supporting cable weight.

Host enumeration

For Android, ADB identifies attached authorized devices by serial and transport. For Apple, management is normally handled through MDM for ongoing fleet operations, with Configurator or local tooling used for specific provisioning/service cases. [S3] [S5]

Symptom	Likely class	Check first
Phone charges, no data	Charge-only cable / port mode	Known-good data cable
Random disconnects	Cable/hub/power instability	Hub supply + logs
Many devices disappear together	Shared hub/controller fault	Upstream hub and AC zone
Slow media transfer	Shared USB bandwidth	Controller topology

10. Thermal engineering

Heat is a reliability and battery-lifetime problem. Apple documents a 0-35 C ambient operating range for iPhone/iPad and warns that high temperatures can permanently reduce battery capacity; charging may slow or stop when the device is too warm. [S11] Other manufacturers publish their own limits, which should be checked per model.

Sources of heat

Charging losses, display use, CPU/GPU work, radios and tightly packed devices all add heat. A rack near a window or inside a closed cabinet can run materially hotter than the room. Measure at phone inlet locations, not only at the HVAC thermostat.

Continuous 100% charge is also undesirable for battery aging. Where the manufacturer provides optimized charging or managed battery features, evaluate them. A fleet can be scheduled to maintain adequate charge without forcing every battery to remain at maximum state of charge continuously.

- Ambient intake temperature.
- Rack exhaust temperature.
- Representative battery/device temperature where accessible.
- Fan operation and dust loading.
- Thermal throttling or charging-paused events.

Airflow

Use a predictable front-to-back or bottom-to-top flow. Recirculation is the enemy: if hot exhaust immediately returns to the intake side, adding more fans can increase noise without improving temperature. Baffles or spacing can be more effective than raw fan count.

Design alerts well below absolute device limits so operators have time to react. The correct warning threshold depends on the device model and sensor being monitored.

Engineering checklist

- Baseline temperatures at 25%, 50% and 100% rack occupancy.
- Repeat test with all devices charging from low battery.
- Define warning and critical thresholds.
- Test a failed-fan scenario.
- Keep rack away from direct solar load.

11. Fire/electrical safety and UPS

A phone rack combines many lithium-ion batteries, charging electronics and AC connections. Use commercially rated electrical components, inspect cords and connectors, and follow local electrical/fire rules. OSHA guidance for extension cords emphasizes proper construction, grounding where required, strain relief and inspection; local regulations may impose additional requirements. [S31]

Safety principles

Do not treat consumer power strips as permanent high-density distribution without verifying their ratings and local code suitability. Do not defeat grounding or overcurrent protection. Remove swollen, damaged or unusually hot batteries from service immediately according to manufacturer and local hazardous-waste guidance.

Provide a clearly reachable main disconnect for the rack and an incident procedure for smoke, odor, battery swelling or abnormal heat. Staff should know not to puncture or compress a damaged lithium-ion battery.

- Rated PDU / distribution.
- Cable inspection schedule.
- No blocked ventilation around chargers.
- Smoke detection appropriate to the site.
- Fire response plan aligned to local authority guidance.

UPS strategy

Prioritize graceful operation rather than attempting to run all charging for hours. A practical UPS can keep router, firewall, switch, APs, controller and database online long enough to preserve state and shut down cleanly. Phones already contain batteries and may not need to be on UPS-backed charging circuits unless continuity requirements demand it.

Calculate UPS VAW load from measured equipment and required runtime. Validate actual runtime with a controlled test after installation.

12. Host computers and local controllers

The local controller is the bridge between the physical fleet and the backend. It may host ADB, local media caching, telemetry collectors, a message broker edge client, a web dashboard, or Apple provisioning utilities.

Sizing

CPU requirements are modest for orchestration but can rise sharply if the host transcodes video, performs local AI analysis or handles many concurrent media transfers. RAM should cover the OS, device tooling, database/cache agents and headroom. Use NVMe storage if the host is a media staging point; otherwise keep large media in object storage or a dedicated NAS.

For a 10-25 device pilot, one reliable workstation or mini-PC may be enough. At higher scale, separate the fleet controller from heavy video processing and from the primary database to reduce blast radius.

- At least two network interfaces if you want clean management/device separation.
- Automatic startup of required services after reboot.
- Disk encryption.
- Admin accounts protected by MFA.
- Regular image/configuration backup.

Operating system choice

Android ADB works across Windows, macOS and Linux. Apple Configurator is a macOS tool, while long-term Apple fleet control should be performed with a suitable MDM rather than requiring every device to remain tethered to one Mac.

Choose the host OS your operators can patch and support reliably. Consistency beats novelty.

13. Inventory, labeling, spares and lifecycle

A fleet becomes manageable when physical identity and database identity are never ambiguous. Build inventory before scale.

Minimum asset record

Each device should have an internal asset ID, manufacturer/model, serial/IMEI where appropriate, purchase source/date, warranty status, OS version, storage, battery-health note, rack/slot, assigned creator or pool, management-enrollment status and retirement date. Sensitive identifiers should be access-controlled.

Do not encode creator names or other sensitive information directly into externally visible device labels. Use opaque asset IDs and resolve them in the management system.

- Asset barcode/QR.
- Cable and charger-port mapping.
- Spare status: ready / repair / quarantine / retired.
- Chain-of-custody when handed to an employee.

Lifecycle

Use a standard state machine: Received -> Inspected -> Enrolled -> Available -> Assigned -> Maintenance -> Quarantined -> Retired -> Wiped/Disposed. Every transition should be timestamped and attributable.

Maintain spare capacity. A fleet that operates at 100% device utilization has no room for OS updates, battery replacement or incident quarantine.

PART III - NETWORKING, CONNECTIVITY, PROXIES AND TUNNELS

Networking is where “phone farm” discussions often become confused. Public IP address, local Wi-Fi address, DNS resolver, proxy, VPN tunnel, SIM carrier and device identity are different concepts. This part separates them and shows how to design reliable, auditable egress without using the network to misrepresent identity or defeat enforcement.

14. Network reference architecture

The recommended baseline is a business firewall/router, managed switching, one or more centrally managed access points, separate VLANs/SSIDs for fleet devices and management, and explicit DNS/egress policy.

Network Topology

Segmentation keeps device traffic, management and storage independently controllable

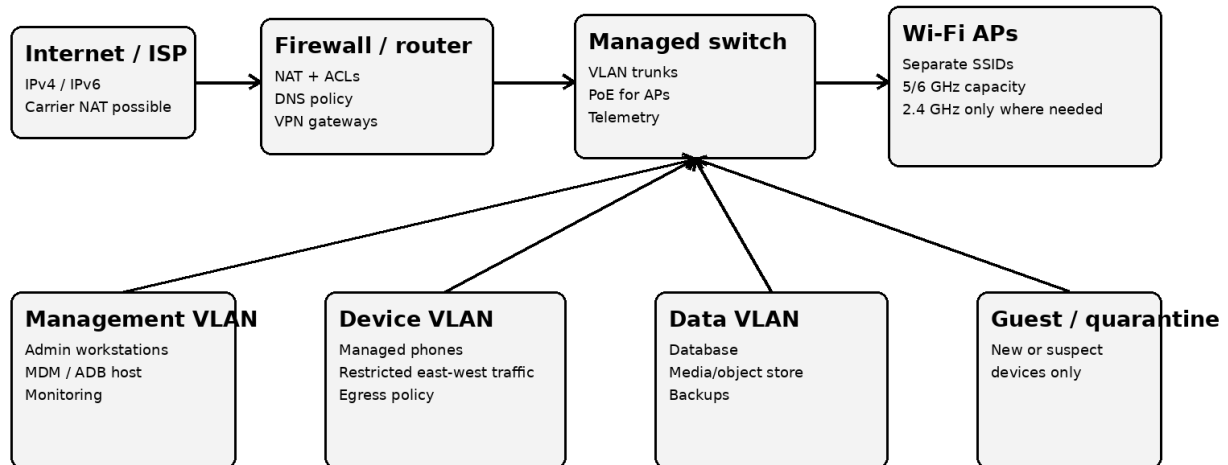


Figure 14-1. Segmented network reference.

Zones

The management zone contains administrator workstations, MDM connectors, ADB hosts and monitoring. The device zone contains phones. The data zone contains database, media store and backups. A quarantine/guest zone can isolate new or suspect devices. ACLs should permit only the flows required between zones.

Segmentation reduces the impact of a compromised phone or stolen employee credential. It also makes troubleshooting easier because you know which traffic class a packet belongs to.

- Management VLAN.
- Device VLAN(s).
- Data/storage VLAN.
- Guest/quarantine VLAN.
- Optional VoIP/office user VLAN separate from the farm.

Internet edge

The firewall should log high-level connection and security events without collecting more personal data than necessary. Where a business VPN gateway or proxy is used, document which devices/apps use it and why.

Prefer deterministic configuration to ad-hoc per-phone settings. Configuration drift is one of the main reasons ten devices behave differently despite looking identical.

15. Wi-Fi radio and capacity planning

Phone density is a radio-capacity problem, not merely a “strong signal” problem. High-density wireless design requires enough airtime, channel planning and AP placement for the concurrent workload. Vendor high-density design guides emphasize planning client density and bandwidth rather than relying on maximum advertised AP client counts.

2.4 vs 5/6 GHz

2.4 GHz travels farther but has less clean spectrum and more interference. 5 GHz provides more channels and is usually preferable for dense phone racks when devices and region support it. 6 GHz can add capacity on compatible devices and APs but has different propagation and regulatory constraints.

Do not put a rack of many phones directly against one AP and assume proximity guarantees capacity. Very strong signals can coexist with contention. Use multiple APs only when channel planning supports it; simply adding APs on overlapping channels can make performance worse.

- Measure channel utilization and retry rates.
- Use wired Ethernet backhaul for APs.
- Disable legacy rates/features only after confirming device compatibility.
- Set reasonable transmit power rather than maximum everywhere.
- Perform throughput and latency tests from representative rack positions.

SSID design

Keep SSID count low because management overhead consumes airtime. Separate networks by security/policy need, not by creating one SSID per creator. VLAN assignment can provide segmentation behind a smaller number of SSIDs.

If using ADB wireless debugging, the workstation and device must meet the applicable pairing/network requirements. Android 11+ supports wireless debugging with pairing; use it on trusted networks only. [S3]

Metric	What it indicates	Typical response
Channel utilization	Airtime pressure	Reduce load / add properly planned capacity
Retry rate	Interference / weak link	RF survey, power/channel tuning
RSSI/SNR	Link quality	AP placement / obstacles
Latency/jitter	Congestion or WAN issue	Separate LAN vs WAN tests
Roaming events	AP handoff behavior	Tune only if devices actually move

16. Ethernet, switches and VLANs

All infrastructure that can be wired should generally be wired. APs, controller hosts, storage and servers should use Ethernet so wireless airtime is reserved for phones.

Switching

A managed switch allows VLAN trunks, per-port statistics, PoE power for APs, and port isolation where useful. Maintain a port map so a failed AP or controller can be found quickly.

For larger installations, redundant uplinks and a second switch may be justified, but complexity should follow actual uptime requirements.

- Label patch cables at both ends.
- Use separate management access for switches/APs.
- Back up switch configuration.
- Disable unused physical ports or place them in a restricted VLAN.

VLAN/ACL design

A VLAN is a Layer-2 segmentation mechanism; it is not security by itself. Inter-VLAN routing rules on the firewall or Layer-3 switch enforce which systems can communicate. A device VLAN typically needs DNS, NTP, internet access and access to explicitly approved internal services, not unrestricted access to the management network.

Block unsolicited device-to-device traffic unless the workflow requires it. This limits lateral movement if one handset is compromised.

17. IPv4, IPv6, NAT and CGNAT

Understanding address translation prevents mistaken assumptions about “one IP per phone.” A phone normally receives a private local IP address from DHCP. The router may translate many private IPv4 addresses into one or several public IPv4 addresses using NAT. A mobile carrier may perform another layer of carrier-grade NAT (CGNAT), meaning many subscribers share public IPv4 space.

IPv4/NAT

Private IPv4 addresses such as RFC1918 ranges are not globally routable. The edge router keeps connection state so return traffic is mapped back to the correct phone. NAT changes addressing; it does not inherently anonymize application identity.

Port/address translation also means a platform may see the same public egress IP for many legitimate devices behind a business connection. Conversely, the same phone may appear behind different public addresses as networks change.

CGNAT and mobile networks

CGNAT performs NAT inside the carrier network at large scale. This is common where public IPv4 addresses are scarce. IPv6 can provide globally routable addresses without the same NAT requirement, though firewalls still control inbound/outbound connectivity. [S18]

Do not design identity assumptions around IPv4 alone. Modern applications can use IPv4 and IPv6 depending on network and DNS responses.

- Record whether each segment has IPv6 enabled.
- Test application behavior over IPv4 and IPv6.
- When troubleshooting “wrong IP,” identify the local IP, router WAN IP and externally observed IP separately.

18. DNS and resolver design

DNS converts hostnames into addresses and is part of both reliability and privacy. Traditional DNS can be visible to the local network; DNS over HTTPS (DoH) and DNS over TLS (DoT) encrypt queries between client and resolver. [S17]

Resolver choices

A business can use the firewall as a forwarding resolver, an internal resolver, or a managed external resolver. The goal is consistent policy, fast failure diagnosis and appropriate logging. Do not mix several ad-hoc per-phone resolver settings unless there is a documented requirement.

If a VPN is used, decide whether DNS should also traverse the tunnel. Split DNS may be necessary for internal names. DNS leaks are not merely a “privacy hack” topic; mismatched routing can break corporate resources and make incident analysis confusing.

- Monitor resolver latency and failure rate.
- Use DNSSEC-validating resolvers where suitable.
- Keep internal hostnames out of public DNS.
- Minimize query logging retention to operational/security needs.

Failure diagnosis

Test in layers: can the phone reach the default gateway, can it reach a known IP, can it resolve a hostname, can it establish TLS to the target, and can the application authenticate? This avoids blaming “the proxy” for a DNS or certificate problem.

19. Forward proxies and proxy protocols

A forward proxy is an intermediary that makes outbound requests on behalf of clients. For HTTP(S), a proxy may forward requests directly or use the CONNECT method to create a TCP tunnel for TLS traffic. [S14] A proxy is different from a VPN: a proxy is usually application/protocol-aware, while a VPN normally routes IP packets through a tunnel.

Proxy, VPN and Egress Concepts

Networking tools are useful for security and segmentation; they do not legitimize identity deception

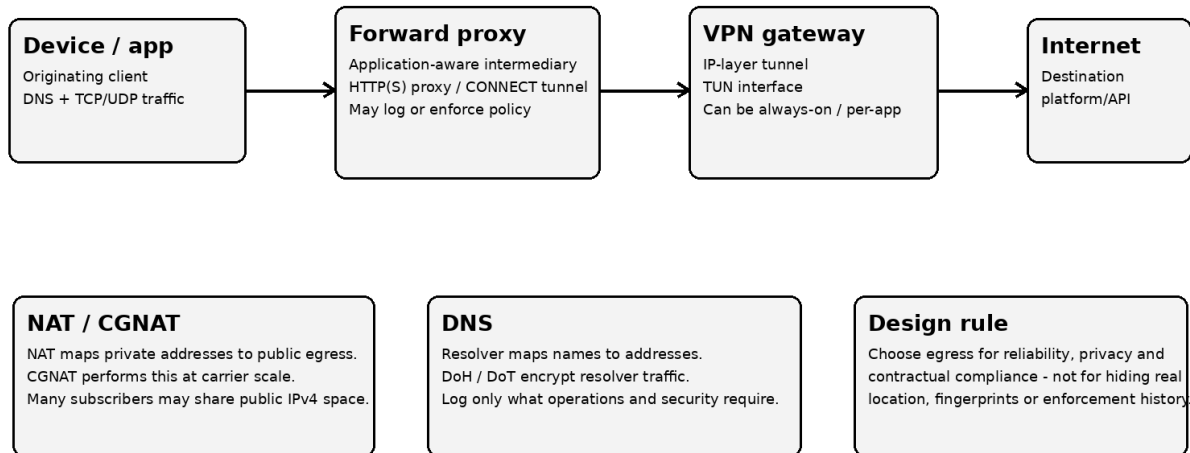


Figure 19-1. Proxy, VPN, NAT and DNS are separate layers.

Why businesses use forward proxies

Legitimate uses include outbound policy enforcement, malware filtering, controlled egress, audit logging, caching in some environments, and allowing specific internal applications to reach the internet through a known gateway.

On mobile devices, proxy support varies by application and OS. A Wi-Fi proxy configured at the network level may not affect every protocol. A VPN-based egress design is often more comprehensive when the requirement is “all traffic from these apps/devices follows this path.”

- HTTP proxy: understands HTTP semantics.
- CONNECT: creates a tunnel commonly used for HTTPS through an HTTP proxy.
- SOCKS-style proxy: relays TCP and sometimes UDP without understanding HTTP content.
- Transparent/intercepting proxy: network redirects traffic; introduces certificate/privacy complexity for TLS inspection.

Proxy provenance and risk

Marketing terms such as “datacenter,” “residential” and “mobile” refer to how the public egress address is sourced. They do not create a compliance exemption. Third-party proxy providers introduce security, privacy, availability and contractual risks because they may observe metadata, alter routes or lose access to address pools.

This manual does not recommend proxy rotation patterns or provider selection for avoiding platform detection. If an agency needs a proxy, choose it for an articulated business purpose such as security, regional testing with authorization, or controlled corporate egress, and perform legal/provider due diligence.

Engineering checklist

- Document why a proxy is needed.
- Confirm which apps/protocols actually use it.
- Test TLS/certificate behavior.
- Measure latency and failure rate.
- Define what the provider can log and retain.
- Create a rollback path to direct egress.

20. VPNs, tunnels and per-app routing

A VPN creates a logical network tunnel between device and gateway. On Android, VpnService creates a virtual TUN interface, allowing a VPN application to read outgoing IP packets and return incoming packets through the interface. Android supports always-on VPN and per-app allow/disallow lists. [S4]

Use cases

VPNs are appropriate for securing traffic over untrusted networks, reaching private corporate services, isolating a fleet behind managed gateways, and applying consistent egress/security policy. WireGuard is one modern protocol designed as a secure network tunnel; enterprise deployments may also use IPsec/IKEv2 or vendor clients. [S19]

Per-app routing can keep only managed applications inside a corporate tunnel while leaving OS update or other traffic direct, or it can do the reverse depending on policy. Test carefully because platform apps may depend on push services, CDNs and media endpoints outside a simple hostname list.

- Always-on mode for high-assurance corporate access.
- Kill/block mode if non-VPN traffic must not bypass policy.
- Per-app lists to scope tunnel use.
- Gateway redundancy for production.
- Key rotation and device revocation.

What a VPN does not do

A VPN changes the network path and may change the public IP address observed by a service. It does not change the account's history, app telemetry, device security state, content, user behavior or the platform's rules. Do not treat it as a method for falsifying identity or location.

21. Cellular, SIM and eSIM concepts

Cellular connectivity can provide redundancy or legitimate mobile testing, but it adds carrier contracts, SIM inventory, signal variation and CGNAT behavior. A mobile plan should be procured and used in accordance with carrier terms and local identity/registration laws.

Physical SIM vs eSIM

Physical SIMs are easy to move between compatible devices but create a physical inventory problem. eSIM can simplify provisioning on supported carriers/devices but may require carrier portals or activation workflows. In either case, map the subscription to the device and cost center in inventory.

Do not confuse the subscriber identity on a SIM with the identity of a social account. Keep telecom records and platform credentials in separate systems.

- Carrier name/plan.
- SIM/eSIM identifier stored with restricted access.
- Assigned device.
- Monthly data cap and roaming rules.
- APN/VPN requirements.
- Suspension/termination procedure.

Redundancy

For business continuity, a cellular router can provide WAN failover for the site. This is often easier to manage than giving every phone its own cellular plan. Individual phone cellular service should exist only where the workflow needs it.

22. Egress observability and troubleshooting

Operators need to answer: which path did this device use, which DNS resolver answered, what public IP was observed, did the VPN/proxy work, how much latency was introduced, and where did the request fail? Build observability for those questions without recording sensitive content unnecessarily.

Telemetry

Collect device ID, network profile, SSID, gateway reachability, resolver status, tunnel status, public egress test result, latency, packet loss and timestamp. Avoid storing credentials, message contents or unnecessary browsing history in network logs.

Use synthetic probes to your own endpoints for health checks. Do not continuously probe third-party platforms in a way that creates abusive load.

- DHCP lease and VLAN.
- VPN/proxy state.
- DNS response time.
- WAN packet loss.
- Gateway CPU/connection utilization.
- AP client count/channel utilization.

Troubleshooting ladder

Always isolate layers: device hardware -> Wi-Fi association -> DHCP -> gateway -> DNS -> tunnel/proxy -> TLS -> application authentication -> platform/API response. A documented ladder reduces random configuration changes that create new faults.

23. Network failure modes

Most fleet networking incidents are ordinary infrastructure failures: overloaded Wi-Fi, dead cables, DNS outages, expired tunnel credentials, saturated WAN links or misapplied ACLs.

Common scenarios

If all devices on one rack fail simultaneously, look for shared AP, switch, VLAN, DHCP or power dependencies. If only one application fails but other internet access works, look above the IP layer. If a subset of devices intermittently disconnects, inspect RF utilization and client-specific power-saving behavior.

When changing proxy/VPN configuration, deploy to a small ring first. A bad egress policy can strand the entire fleet from MDM or management services.

Resilience patterns

Dual WAN, redundant DNS resolvers, spare APs, configuration backups and staged changes provide more benefit than exotic networking. Keep a direct, secure management path so a failed content egress tunnel does not remove your ability to manage the device.

- Configuration versioning.
- Change tickets with rollback.
- Canary device/rack.
- Out-of-band admin path where justified.
- Post-incident review.

Failure	Blast radius clue	First containment
AP failure	One RF cell/rack	Move clients / replace AP
DHCP failure	New joins fail	Restore DHCP / lease service
DNS failure	IP works, names fail	Fail over resolver
VPN gateway failure	Tunnel users only	Fail over or controlled bypass
WAN outage	Whole site	Secondary ISP/cellular failover

PART IV - DEVICE FLEET MANAGEMENT

Fleet management should make devices boring: repeatable enrollment, predictable settings, known application versions, controlled access and fast recovery. Apple and Android require different approaches.

24. Apple fleet architecture

Apple's enterprise deployment model centers on organization ownership, Automated Device Enrollment where available, supervision and Mobile Device Management (MDM). Apple Platform Deployment documents the lifecycle from planning and enrollment through configuration and ongoing management. [S5]

Apple Fleet Control Flow

Supervision + MDM for organization-owned iPhone/iPad fleets

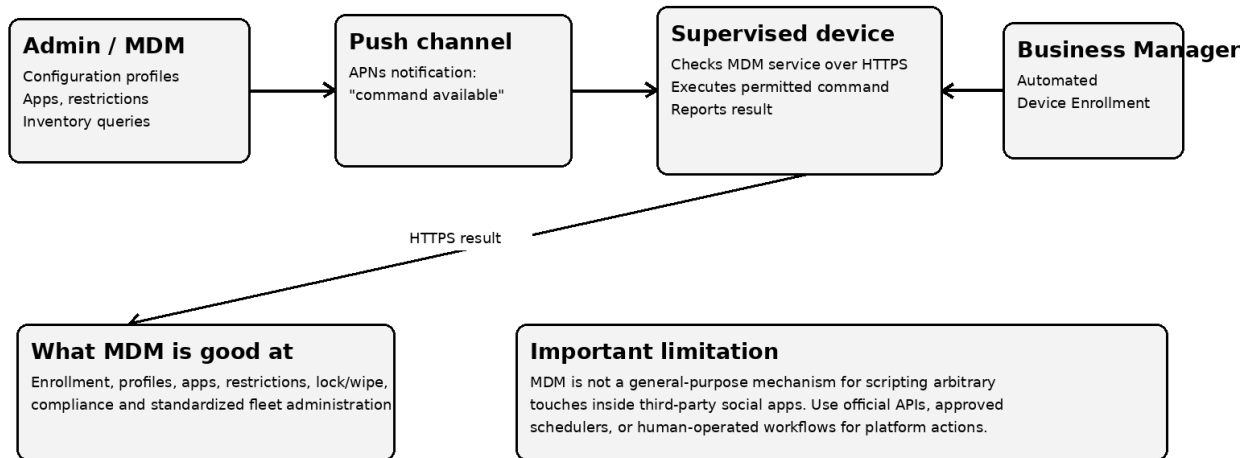


Figure 24-1. Simplified Apple MDM control flow.

Core components

Apple Business Manager (or the appropriate Apple organizational service) establishes organization ownership and can assign devices to an MDM service. Supervision provides additional management controls for organization-owned devices. The MDM service sends configuration profiles, app assignments and commands while devices communicate with Apple push infrastructure and the MDM service.

Use a commercial or internally operated MDM that explicitly supports the commands, restrictions and app deployment you require. The phone rack itself should not depend on one operator manually configuring every iPhone.

- Organization account and administrative roles.
- Device enrollment/assignment.
- Supervision.
- MDM server/service.
- APNs connectivity.
- Managed app distribution.

Design principle

MDM is the administrative plane. It is not a generic remote-control framework for simulating user gestures in third-party applications. Keep publishing workflows separate.

25. Supervision, enrollment and MDM

Supervision signals that an iPhone/iPad is organization-controlled and unlocks additional management capabilities. Automated Device Enrollment can supervise eligible organization-owned devices during setup; Apple Configurator can be used in certain manual preparation workflows. [S5]

Enrollment flow

At acquisition, record the serial and organization ownership. Assign the device to the intended MDM, boot/reset it, complete Automated Device Enrollment, verify supervision, apply baseline profiles, install managed applications, test connectivity, then mark the device Ready in inventory.

MDM commands typically use push notification to tell the device that a command is available; the device then contacts the management service over HTTPS, executes supported actions and reports results. [S6][S7]

- Wi-Fi/certificate profiles.
- Passcode and security policy.
- Managed apps.
- Restrictions appropriate to role.
- Lost mode/lock/wipe capability where supported.
- Inventory and compliance reporting.

Recovery

If a supervised device is lost or compromised, use the MDM's supported lock/wipe workflow and revoke relevant credentials. If it is reassigned, wipe and re-enroll rather than merely signing out of one social application.

26. Android Enterprise architecture

Android Enterprise provides management modes for personally owned and company-owned devices. For a phone farm, company-owned fully managed devices are normally the cleanest model because the organization can manage the complete work-only device. Dedicated devices are a subset that can be locked to a small application set. [S1]

Android Fleet Control Flow

Android Enterprise for policy; ADB for authorized engineering and diagnostics

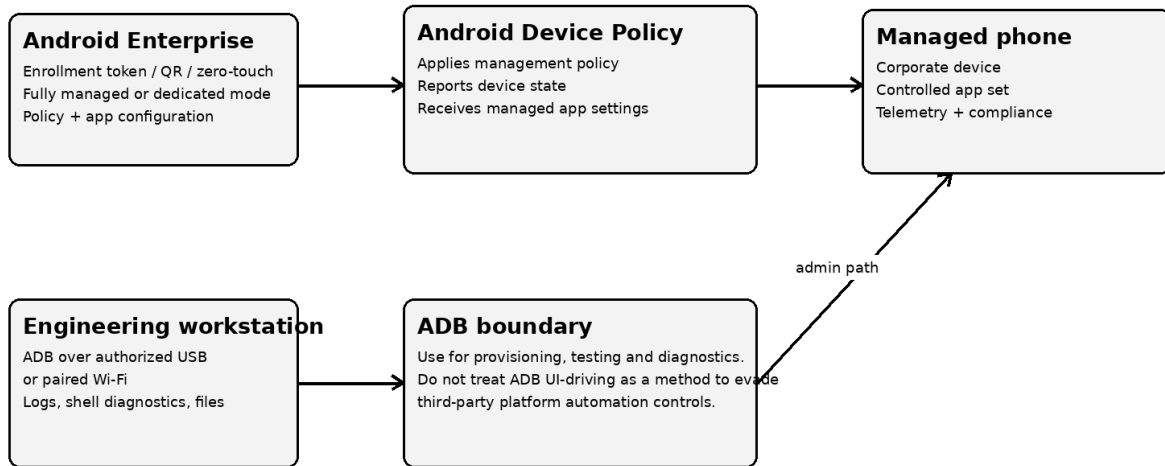


Figure 26-1. Android Enterprise plus authorized ADB engineering.

Android Management API

Google's Android Management API uses Android Device Policy on the device to receive and enforce enterprise policies. Provisioning creates a device object bound to the enterprise. Enrollment tokens determine ownership/management mode; company-owned devices can use fully managed configurations. [S1]

Policies can control application installation, permissions, restrictions, networking settings, system updates and compliance behavior depending on Android version and management mode.

- Enterprise resource.
- Enrollment tokens.
- Policies.
- Device objects and state.
- Managed Google Play application policy.

Dedicated vs fully managed

Dedicated mode is suitable when the phone should expose only a controlled application set. If operators need several social apps, browser, camera, authenticator and internal tools, a normal fully managed configuration may be more practical than an overly restrictive kiosk.

27. Android provisioning and policy

Google documents QR, zero-touch, NFC and other provisioning methods. For company-owned work-only devices, full management allows the enterprise to manage all applications and enforce broad policy. [S1]

Pilot provisioning

For a small pilot, QR enrollment is straightforward: factory-reset the corporate device, enter the Android provisioning flow, scan the organization-generated enrollment QR, allow Android Device Policy to complete setup, then verify policy/application state. At scale, eligible zero-touch procurement reduces hands-on setup.

Enrollment tokens should be short-lived where practical and protected as administrative secrets because possession can allow a device to enroll into the enterprise configuration.

- Baseline Wi-Fi or setup network.
- Enrollment token handling.
- Policy assignment.
- Required apps.
- VPN client if required.
- Compliance validation.

Policy rings

Use Stable, Canary and Quarantine policies. Canary receives new OS/app/policy changes first. Stable is production. Quarantine removes unnecessary access while preserving enough management connectivity for investigation.

28. ADB for authorized engineering

Android Debug Bridge (ADB) is Google's command-line bridge between a development workstation and Android devices. It can communicate over USB and, on modern Android, paired wireless debugging. [S3] It is appropriate for provisioning support, logs, diagnostics and controlled testing on devices you own or are authorized to administer.

Architecture

ADB consists of a client, local server and device daemon. A workstation can enumerate authorized devices, open a shell, collect logs and transfer files. USB is usually the most predictable for a rack; wireless debugging reduces cables but depends on trusted Wi-Fi and pairing state.

Use a dedicated engineering account and workstation. Disable debugging on production devices if it is not needed, or restrict the workstation/network that can reach it.

- Inventory verification.
- Log collection.
- File push/pull for internal test assets.
- Package/version diagnostics.
- Network diagnostics.

Boundary

ADB can technically generate input events and interact with applications. This manual does not provide a workflow for using ADB to automate third-party social UIs where automation is prohibited or to make automated behavior appear human. Use supported APIs or human operation for those actions.

29. App distribution and configuration

A managed fleet needs predictable application versions. Install apps through MDM/managed app channels where available, define which apps are required, and avoid ad-hoc APK or IPA distribution unless you own the software and have an authorized enterprise development path.

Application catalog

Maintain a catalog with app name, package/bundle ID, platform, source, minimum version, required permissions, owner, data classification and update ring. Social apps, password manager, authenticator, VPN client, internal task app and device-health agent should each have an owner.

Permissions should be the minimum needed. For example, a publishing app may need photo-library access but not contacts. Review permissions after major app updates.

- Required vs optional.
- Version and update policy.
- Permissions.
- Data stored locally.
- SSO support.
- Uninstall/retirement behavior.

Internal companion app

Consider a small internal “operator companion” app that shows device ID, assigned creator/account, task queue, approved asset links, check-in status and incident button. It should not store social-account passwords and should not covertly drive other applications.

30. OS updates and maintenance rings

Ignoring OS updates creates security debt; updating every device immediately creates operational risk. Use rings.

Rings

Canary devices receive updates first and run representative workloads. After a defined observation window, a larger pilot ring updates, then production. Critical security fixes may justify accelerated rollout after compatibility testing.

MDM or Android Enterprise update controls can help schedule or defer updates within platform-supported limits. Do not leave devices indefinitely frozen on unsupported releases.

- Canary: 1-5% of fleet.
- Pilot: 10-20%.
- Production: remaining devices.
- Emergency: accelerated security rollout with explicit approval.

Maintenance window

Drain publishing tasks from the device, confirm battery/power, install update, reboot, verify management heartbeat, verify required apps and network, then return to Ready. Record the new build version automatically where possible.

31. Health telemetry and remote actions

A fleet dashboard should answer “what is healthy right now?” without requiring someone to walk down the rack.

Health model

At minimum track last heartbeat, OS/app version, storage free, battery/charging state, management compliance, network profile, last task, last error and current assignment. Temperature data availability varies by OS/device; use rack sensors even when device-level telemetry is limited.

Status should be derived, not subjective: Ready, Busy, Maintenance, Degraded, Quarantined, Offline.

- Heartbeat age.
- Charging state.
- Free storage.
- Wi-Fi/VPN state.
- Required app versions.
- MDM/Enterprise compliance.
- Task error rate.

Remote actions

Administrative actions can include lock, wipe, restart where supported, reapply policy, request inventory, rotate internal app tokens, or move a device to quarantine. High-impact actions should require stronger authorization and be written to the audit log.

PART V - SOFTWARE CONTROL PLANE

The coordinator makes the phone farm a system rather than a pile of devices. It should model devices, creators, accounts, assets, approvals, tasks, results, networks and employees as separate entities with an auditable workflow.

32. Control-plane architecture

A practical control plane consists of an API service, relational database, object/media storage, task queue or broker, device/fleet connectors, authentication/RBAC, metrics/logging and an operator web interface.

Reference Architecture

A compliant, observable creator-traffic operations platform

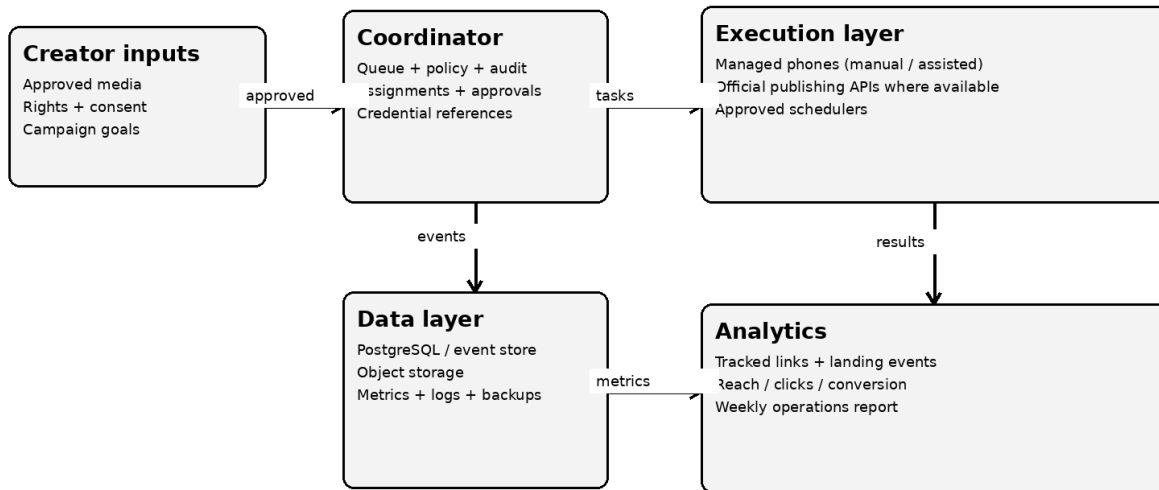


Figure 32-1. Control plane and execution layers.

Service boundaries

The API service owns business rules. The relational database stores durable entities and task state. Object storage holds media and large artifacts. A broker or queue delivers asynchronous work. Device connectors translate generic management intents into Apple MDM, Android Enterprise, internal companion-app or other supported actions.

Keep platform publishing connectors separate from device-management connectors. This prevents a social API outage from breaking fleet administration and makes authorization boundaries clearer.

- API/auth service.
- PostgreSQL or equivalent relational store.
- Object storage.
- Queue/broker.
- Fleet connectors.
- Publishing connectors.
- Analytics/event pipeline.
- Web dashboard.

Deployment

For a pilot, these can run on one secured server using containers. For production, separate stateful services, backups and secrets. Avoid unnecessary microservices: operational complexity grows faster than device count.

33. Device-to-coordinator communications

There are two broad channels: enterprise management channels provided by Apple/Google, and a custom internal agent/companion channel controlled by the agency. Do not reinvent MDM for functions MDM already performs.

HTTPS model

An internal companion app can periodically call an HTTPS endpoint to register heartbeat, fetch permitted task metadata and report completion. Use short-lived OAuth-style tokens or device certificates rather than embedding a shared static API key in every phone.

For near-real-time updates, WebSocket or server-sent events can be added, but simple polling is often easier to operate at 10-100 devices. Poll intervals should use jitter so all devices do not hit the server simultaneously.

- TLS for all remote traffic.
- Unique device identity.
- Token rotation/revocation.
- Idempotent task receipt.
- Clock synchronization.
- Backoff on failure.

Message broker

MQTT is a lightweight publish/subscribe protocol widely used for constrained devices and IoT-style messaging. [S20] A companion app could publish health telemetry and subscribe to a narrow command topic, but do not expose a broker directly to the public internet without strong authentication, authorization and TLS.

A queue/broker is transport, not the source of truth. Durable task state belongs in the database.

34. Task queue and state machine

Every action should move through explicit states so operators know whether it is waiting, running, verified or failed. Silent retries hide problems and can create duplicate posts.

Task Lifecycle

Every publishing or device action should have an accountable state transition

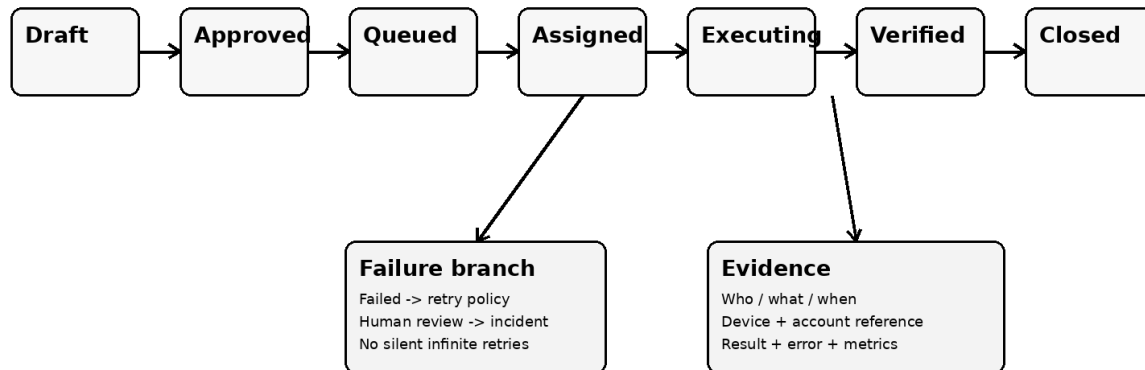


Figure 34-1. Auditable task lifecycle.

Reference states

Draft: not yet approved. Approved: content/action permitted. Queued: eligible for scheduling. Assigned: bound to executor/device/operator. Executing: in progress. Verified: expected result observed. Closed: complete and immutable except for annotations. Failed tasks enter a review/retry branch with a reason.

Use idempotency keys for API publishing so a network retry does not create a second post. Human tasks should require the operator to record the platform post ID or a verification artifact before closure.

- Maximum retry count.
- Retryable vs terminal error classes.
- Deadline/SLA.
- Priority.
- Executor type.
- Required approval IDs.
- Verification method.

Scheduling

Scheduler logic should consider platform/account restrictions, operator shift, device readiness, campaign windows and content approvals. It should not create prohibited spam by blindly maximizing throughput.

35. Database and entity model

The database should model real business entities instead of embedding everything in a spreadsheet row. This supports permissions, history and reporting.

Core Data Model

Keep identities, assets, devices, accounts and tasks as separate entities

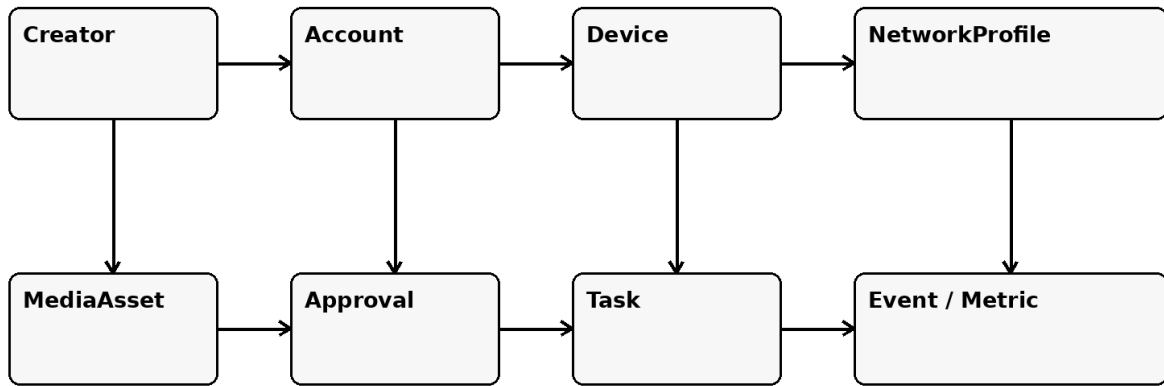


Figure 35-1. Core entity separation.

Core entities

Creator represents the contracted person/business. Account represents one platform account and stores ownership/authorization metadata, not plaintext credentials. Device represents a physical asset. NetworkProfile describes approved egress configuration. MediaAsset stores immutable asset metadata and storage URI. Approval links a person to a permitted action/asset. Task describes work. Event/Metric stores observed outcomes.

Use stable UUIDs internally. Human-readable labels can change; primary identity should not.

- Creator.
- PlatformAccount.
- Device.
- NetworkProfile.
- MediaAsset / AssetVersion.
- Approval.
- Task / TaskAttempt.
- Operator.
- AuditEvent.
- MetricEvent.

History

Use append-only audit records for security-relevant changes. For mutable business records, store updated_at and updated_by; for especially sensitive fields keep version history or event-sourced change records.

36. Object/media storage

Creator media is valuable and sensitive. Centralize it in access-controlled object storage rather than uncontrolled phone camera rolls and employee downloads.

Storage model

Use immutable source assets plus derived versions. Store SHA-256 or another strong checksum for integrity, creator ID, usage authorization, source, creation/import timestamp, classification and retention/expiration. A derived version should point back to its source and transformation recipe.

Phones should download only the assets required for assigned tasks, ideally through time-limited signed links. Clear local staging areas on reassignment/offboarding where technically possible.

- Server-side encryption.
- Restricted buckets/containers.
- Versioning where useful.
- Lifecycle/retention rules.
- Backup/replication.
- Access logging.

Content safety

Only store adult creator content for verified adults with documented authorization and applicable consent/releases. Build a takedown/revocation process that can locate every derived copy from an asset ID.

37. API design

The internal API should expose business intentions, not raw device-control primitives. “Assign approved asset X to account Y” is safer than “tap coordinate 400,900.”

Resource examples

Use REST or another consistent RPC model with resources such as /devices, /accounts, /assets, /approvals, /tasks, /events and /metrics. Every write should authenticate the caller and enforce role/creator scope.

Use request IDs and idempotency keys. Return machine-readable error codes. Record actor, source IP/device, target object and result in audit logs for security-relevant writes.

- GET /devices/{id}.
- POST /tasks.
- POST /tasks/{id}/approve.
- POST /tasks/{id}/assign.
- POST /tasks/{id}/result.
- GET /accounts/{id}/metrics.

Versioning

Version the API contract before external clients depend on it. Internal rapid changes are easier if the companion app and backend can tolerate one previous schema version during rolling updates.

38. Messaging and event transport

Asynchronous messaging decouples slow device/network actions from interactive web requests. A user should not keep an HTTP request open while waiting for a phone task to finish.

Queue semantics

At-least-once delivery is common and requires idempotent consumers. Exactly-once behavior is usually implemented at the application level using unique keys and database constraints rather than assumed from the broker.

Use separate queues/topics for device commands, publishing jobs, analytics ingestion and low-priority maintenance so one workload cannot starve another.

- Dead-letter queue.
- Retry delay/backoff.
- Maximum age.
- Consumer concurrency.
- Per-account/device serialization where necessary.

MQTT topic example

For telemetry, a namespace such as `fleet/{device_id}/health` and `fleet/{device_id}/events` is easy to authorize. Commands should be narrow and typed. Do not create a universal remote-shell topic in a production broker.

39. Metrics, logs and traces

Prometheus is a common metrics and alerting system built around time-series data and label dimensions. [S21] Whether using Prometheus or another stack, distinguish metrics, logs and traces.

Metrics

Metrics answer aggregate questions: online device count, task success ratio, queue depth, API latency, Wi-Fi loss, VPN tunnel availability. Keep label cardinality controlled; do not use a unique post URL as a Prometheus label if it creates millions of series.

Logs answer event details: device enrolled, task failed, credential reference rotated. Traces connect calls across services when debugging distributed workflows.

- Fleet online %.
- Task success/failure by executor.
- Queue age P50/P95.
- Publishing connector error rate.
- Database latency.
- Storage capacity.
- Network health.

Alerting

Alerts should be actionable: ">20% of devices in rack B missed three heartbeats" is useful; "CPU 71%" may not be. Add runbook links to alerts.

40. Credential and secret handling

Credentials are one of the highest-value assets in an OFM system. Do not store platform passwords, session cookies, MFA seeds or API keys in source code, shared spreadsheets, task descriptions or chat channels.

Vault model

Use a password/secret manager that supports per-user access, audit logs, MFA and revocation. The application database stores a reference such as `vault://creator/123/platform/x` rather than the secret value. Backend services receive only the secret they need, ideally for a limited time.

Where a platform supports delegated access, teams/business accounts or OAuth, prefer that over sharing the creator's primary password.

- Unique operator accounts.
- MFA/hardware keys for administrators.
- No shared root credential.
- Recovery codes stored securely.
- Immediate revocation on offboarding.
- Periodic access review.

Session management

Social apps may maintain long-lived sessions on phones. Treat the device itself as a credential-bearing asset: lock it, encrypt it, inventory it, and wipe/revoke on loss or reassignment.

41. Availability, backup and disaster recovery

The control plane should survive ordinary failures without losing authorization records, task history or media.

Backup scope

Back up relational databases, configuration, audit records, object-storage metadata and essential media according to business/legal requirements. Secrets should be backed up through the vault's supported recovery mechanism, not exported into general backups.

Test restore. A backup that has never been restored is an assumption, not a recovery plan.

- Define RPO: acceptable data loss window.
- Define RTO: acceptable service recovery time.
- Keep at least one backup copy isolated from primary credentials.
- Document dependency order: identity -> database -> object store -> queue -> API -> workers.

Degraded operation

If the coordinator is down, operators should have a clear rule: pause new publishing or use an approved manual fallback with later reconciliation. Do not create a shadow spreadsheet that permanently bypasses audit and approval controls.

PART VI - OFM OPERATIONS, CONTENT AND ANALYTICS

The technical fleet exists to support a controlled creator-content workflow. This part focuses on authorization, media, publishing, employees and measurement.

42. Creator onboarding and authorization

Before an account or device is assigned, the creator relationship should be documented and operationally represented in the system.

Onboarding packet

Record legal/business identity as required by the agency, contract reference, platform account ownership, recovery-channel ownership, allowed content categories, geographic/commercial restrictions, approved employees, revenue attribution identifiers and exit procedure. Sensitive identity documents require stricter access than routine content metadata.

The creator should know which agency systems and people will access which accounts. Authorization should be revocable.

- Contract/authorization.
- Account inventory.
- Recovery contacts.
- Content rights/consent.
- Brand rules.
- Escalation contacts.
- Offboarding instruction.

Technical onboarding

Create creator and account records, issue vault entries, assign least-privilege roles, import only approved media, configure tracking links, run one controlled publishing test and verify analytics before scaling.

43. Media ingest and rights metadata

Every media asset should enter through a controlled ingest point instead of being copied through random employee devices.

Content Operations Pipeline

Rights and approval precede distribution; performance data feeds the next creative decision

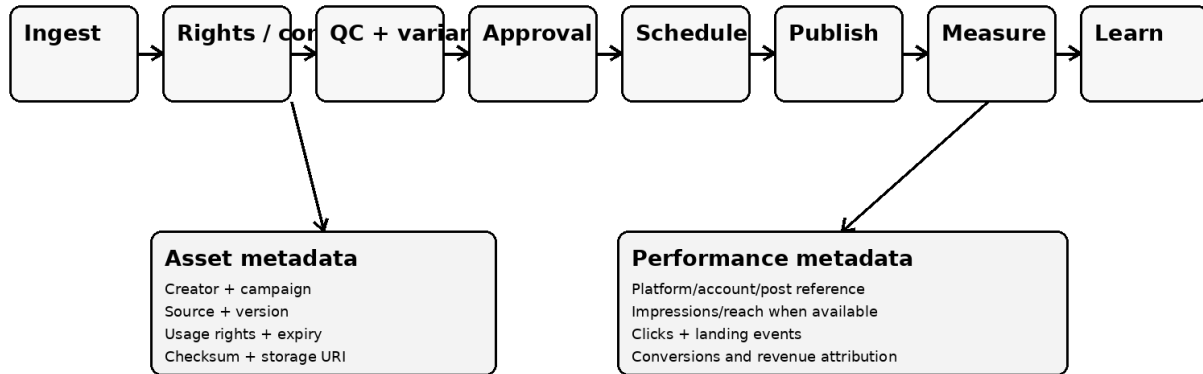


Figure 43-1. Rights-aware content pipeline.

Ingest

When an asset is received, calculate a checksum, store the original immutably, attach creator/campaign ownership, record source and rights scope, and create a QC status. The ingestion system should not silently strip or rewrite evidence relevant to rights and provenance.

If the agency performs edits, retain an edit lineage so the approved published variant can be traced to the source.

- Asset UUID.
- Checksum.
- Creator/campaign.
- Source.
- Rights/consent reference.
- Import timestamp.
- Classification.
- Retention/expiry.

Sensitive content

Adult material requires strict access control. Avoid unrestricted local synchronization. Use role-based access and watermarking for internal review only if it does not alter source evidence and the creator approves the workflow.

44. Content processing and variants

Content processing should optimize for platform-supported formats and campaign testing without deceptive manipulation.

Variant generation

Create explicit variants for aspect ratio, safe crops, subtitle/caption burn-in where authorized, cover frame, encoding bitrate and clip length. Store the transformation recipe and version ID so results are reproducible.

Use platform-published media specifications and test upload quality. Avoid “metadata spoofing” intended to make repeated content appear unrelated or to deceive integrity systems; that is different from legitimate transcoding or metadata minimization for privacy.

- 9:16 vertical variant.
- Platform-safe crop.
- Captioned variant.
- Thumbnail/cover.
- Source-preserving archival copy.

Quality control

Automated QC can detect resolution, duration, codec, audio presence, file corruption and duplicate checksum. Human QC should confirm brand, consent, visible private information and platform suitability.

45. Approval workflow

Approval is a gate, not a chat reaction. A task should prove which version was approved, by whom, for which account and time window.

Approval object

Approval should reference immutable asset version, creator/campaign, target platforms/accounts or a defined scope, approver identity, timestamp and expiration/revocation if applicable. A later edit creates a new version requiring new approval unless the policy explicitly allows that transformation.

High-risk or ambiguous content can require two-person review: creator/manager plus compliance/operations.

- Asset/version.
- Scope.
- Approver.
- Timestamp.
- Expiry.
- Notes/restrictions.

Revocation

If a creator withdraws authorization, the system should be able to identify queued tasks and previously published references associated with the revoked asset so operators can stop future use and execute contractual/platform removal procedures.

46. Publishing paths and official APIs

Publishing should use the most supportable path available: official API, approved scheduler/integration, or human-operated managed phone. The coordinator records all three the same way.

Official API path

TikTok's Content Posting API allows registered applications to post content on behalf of authorized creators after the app obtains the required product configuration and scope; unaudited clients have restrictions. [S22] This demonstrates the preferred automation pattern: explicit creator authorization, documented API scope and platform-controlled rate/feature constraints.

For other platforms, verify current first-party developer documentation because availability depends on account type, application review and changing product policy.

- OAuth/delegated authorization where supported.
- Rate limiting.
- Idempotency.
- Post/status ID capture.
- Audit of API actor and creator authorization.

Human device path

If an action is mobile-only or API-inaccessible, create an operator task with approved asset, caption guidance and target account. The operator performs the action in the native app and records verification. The coordinator should assist the human, not secretly impersonate the human.

47. TikTok operational notes

TikTok is particularly explicit about integrity and automation. Its current guidelines allow authentic multi-account use but prohibit spam, fake engagement, circumvention and automation used to register or operate accounts in bulk. [S23]

Compliant engineering posture

Use the Content Posting API when the desired workflow is supported and the application can satisfy authorization/audit requirements. Otherwise use trained operators on authorized creator accounts and keep posting cadence/content strategy inside current platform rules.

Build analytics around content quality and conversion rather than account multiplication. A smaller number of healthy, authorized accounts with measurable attribution is easier to govern than a disposable-account model.

- Do not automate bulk account creation.
- Do not buy/sell or misrepresent account identity.
- Do not manufacture views, likes, comments or follows.
- Do not build location-hiding or enforcement-circumvention mechanisms.

Operational telemetry

Record post ID, publishing path, asset version, timestamp, operator/API application, outcome and available reach/click data. If a post is rejected or account restricted, preserve the platform notice and route it to policy review rather than automatically creating replacements.

48. Reddit, X and Snapchat operational notes

Each platform has its own rules, but authenticity and spam controls are common themes.

Reddit

Reddit requires authentic participation and prohibits spam/content manipulation. [S24] Subreddit-specific rules can be stricter than site-wide rules. Build a subreddit rule register if Reddit is part of the creator strategy; do not assume an action allowed in one community is allowed in another.

Where API access is used, comply with current Reddit Data API terms and rate/access conditions. [S25]

- Record subreddit/community rules and last review date.
- Avoid repetitive cross-posting that violates community spam rules.
- Respect NSFW labeling and content restrictions.

X

X's authenticity rules recognize legitimate third-party management but prohibit coordinated inauthentic behavior, duplicate/substantially similar amplification and ban evasion. [S27] This supports a model where the agency manages authorized creator accounts without creating fake networks to manipulate reach.

Snapchat

Snapchat's deceptive-practices policy prohibits spam, artificially inflated engagement, follower-growth schemes and other deceptive behavior. [S26] Native content workflows should therefore remain human or platform-supported rather than driven by hidden engagement bots.

49. Instagram/Meta operational notes

Meta's products have formal business/developer tooling as well as anti-spam and authenticity rules. Exact publishing capabilities and account requirements change, so the deployment should treat Meta integrations as versioned connectors that are validated against current first-party documentation before release.

Source note

Meta policy/API pages can change frequently and were not used here as a step-by-step integration specification. Validate the exact current first-party endpoint, account type and review requirements at implementation time.

Architecture approach

Prefer supported Meta business/account access and approved publishing integrations where available. Maintain separate account ownership, page/profile/business relationships and authorization records in the control plane.

Avoid building the system around large-scale repetitive posting, synthetic engagement or replacement-account churn. Even if a device can technically perform an action, the business should first ask whether the action is permitted and sustainable.

- Verify current Instagram API publishing requirements.
- Use delegated business roles where available.
- Capture post/media IDs for reconciliation.
- Monitor access-token expiry and permission changes.

Native app tasks

Stories, certain editing features or emerging formats may reach the native app before APIs. Route these to a human task and maintain evidence of the approved asset and target account.

50. OnlyFans boundary and account responsibility

OnlyFans is the monetization endpoint in many OFM funnels, but the social phone farm and the OnlyFans account should be treated as separate systems. Agency assistance does not remove the creator's legal or contractual responsibility for the account. Publicly archived terms and UK parliamentary evidence have historically emphasized creator control/responsibility; verify current OnlyFans Terms directly before implementation. [S28][S29]

Verification warning

OnlyFans terms are especially important to verify at the time of deployment. Some accessible web sources are archived or secondary, so do not rely on this chapter as a current legal opinion.

Access

Do not assume that a social-media operations employee needs OnlyFans credentials. Separate traffic acquisition roles from subscriber/chat or account-administration roles. This reduces both privacy exposure and credential blast radius.

If third-party account assistance is permitted under current terms and creator contract, document the authority, employee scope, authentication method and revocation process.

- Creator retains required ownership/control.
- Agency access explicitly authorized.
- Separate roles and least privilege.
- MFA/recovery ownership documented.
- Offboarding and session revocation tested.

Data separation

Keep public social analytics, creator media and subscriber/private customer data in separate access domains. Do not copy subscriber conversations into a growth dashboard unless there is a documented lawful/business need and appropriate protection.

51. Tracking links and attribution

Without attribution, a phone farm can produce activity without knowing what generates revenue. Use campaign-aware links and first-party landing analytics where lawful.

Link structure

Create a campaign/link ID that resolves to the intended destination. Store creator, platform, account, campaign, asset and optional post reference. The redirect service records only necessary event data and should respect privacy/consent requirements.

UTM parameters are useful for general analytics, but an internal opaque link ID can be easier to rotate or update without exposing internal naming.

- Link ID.
- Creator/campaign.
- Source platform/account.
- Asset/post reference.
- Created/expired dates.
- Destination.

Attribution levels

Deterministic attribution uses a directly observed click and conversion identifier. Probabilistic or modeled attribution estimates influence when deterministic linking is unavailable. Label modeled results clearly; do not present them as exact facts.

Use cohort comparisons and revenue per campaign rather than optimizing solely for impressions.

52. Employee roles and access model

Employees are part of the system. Define roles so a new hire does not automatically receive every creator credential and media library.

Example roles

Fleet technician manages devices/network but does not need creator social passwords. Content operator sees assigned approved assets and target accounts. Growth manager sees analytics and campaign configuration. Security/admin manages access and incidents. Finance sees revenue/cost summaries but not raw sensitive content unless required.

Role-based access should be supplemented by creator/account scope. A content operator may be authorized for Creator A but not Creator B.

- Fleet technician.
- Content operator.
- Growth manager.
- Creator manager.
- Security/admin.
- Finance/analytics.

Joiner/mover/leaver

Provision named accounts, train the employee, issue hardware/keys, review access when role changes, and revoke access immediately at departure. Rotate shared recovery secrets if any existed, and reconcile devices/keys physically returned.

Role	Devices	Creator media	Social credentials	Analytics
Fleet tech	Admin	No/minimal	No	Health only
Content operator	Assigned	Assigned only	Delegated/assigned	Task-level
Growth manager	View	Campaign	Usually no password	Broad
Security admin	Admin	As incident requires	Vault admin only	Audit
Finance	No	No	No	Revenue/cost

53. Daily/weekly operating cadence

A phone farm should be run as an operations function with handoffs, not as continuous improvisation.

Daily

Start with health: offline devices, failed backups, full storage, expired credentials, network alerts and platform notices. Then review the approved publishing queue. End the shift with unresolved incidents and blocked tasks explicitly handed over.

Avoid measuring employees purely by number of posts; that incentivizes spam and hides quality or conversion problems.

- Morning fleet health check.
- Approval/queue review.
- Execute tasks.
- Verify and reconcile post IDs.
- Review alerts/restrictions.
- Shift handoff.

Weekly

Report output, funnel metrics, account health, device/network reliability, policy incidents, top/bottom content, labor usage and next-week experiments. The report should say what is blocked and who owns resolution.

54. Quality control and audit

Quality control verifies content, account and task correctness; audit verifies who changed what.

QC sampling

Randomly sample completed tasks and compare the published post with the approved asset/caption/target. Verify links resolve correctly and attribution metadata is present. Sample failed tasks to ensure error classification is honest rather than simply retried until “success.”

Periodically inventory physical devices against the database and MDM/Android Enterprise records.

- Published asset matches approval.
- Correct creator/account.
- Correct tracking link.
- Post ID captured.
- No unauthorized local asset remains.
- Device assignment matches physical rack.

Audit log

Write immutable or append-only records for login, role changes, credential access, creator authorization changes, task approvals, high-impact device commands and deletions. Retain according to legal/business requirements, not indefinitely by default.

PART VII - SECURITY, COMPLIANCE AND RELIABILITY

The fleet contains devices, credentials and sensitive creator content. Security therefore has to be designed into networking, software and employee workflows.

55. Threat model

A threat model identifies assets, adversaries, attack paths and controls. The most realistic threats are often credential theft, malicious/curious insiders, lost devices, vulnerable phones, exposed admin panels, leaked media and configuration mistakes.

Assets

Creator accounts, social sessions, content, legal authorization records, subscriber/private data where applicable, analytics/revenue, device inventory, network credentials and administrative secrets.

Rank assets by impact and limit who can access them. A fleet technician does not need subscriber data; a growth analyst does not need MDM wipe rights.

- External credential phishing.
- Malware on admin workstation.
- Stolen/unreturned phone.
- Insider media exfiltration.
- Misconfigured public storage.
- Compromised VPN/proxy account.
- Unpatched mobile OS.

Controls

MFA, vaulting, MDM, encryption, segmentation, EDR on admin hosts, patching, least privilege, secure backups, audit logs and employee offboarding cover most high-probability paths better than exotic anti-fraud tricks.

56. Platform restriction and enforcement handling

When a platform restricts a post or account, treat it as a policy/operations incident, not an automatic trigger to create replacement identities.

Workflow

Preserve the notice, account/post ID, timestamp, asset, publishing path and recent changes. Determine whether it is content policy, spam/integrity, security login, copyright, age/consent or an unknown issue. Use the platform's appeal/remediation process where appropriate.

Pause similar queued tasks if the issue may repeat. A single restriction can reveal a broken campaign rule that would otherwise affect dozens of accounts.

- Capture evidence.
- Classify cause.
- Contain similar tasks.
- Appeal/remediate legitimately.
- Update campaign/policy rule.
- Document decision.

Prohibited response pattern

Do not make "ban evasion" a software feature. Automated creation of replacement accounts, identity/fingerprint spoofing, location falsification or rapid proxy/device churn can violate platform rules and create fraud/security risk. TikTok and X explicitly address circumvention/ban evasion in their authenticity rules. [S23][S27]

57. Incident response

Incident response should be predefined because the first minutes of a credential leak or overheating event are not the time to invent authority and evidence rules.

Incident Response Flow

Design for containment and evidence before an incident happens

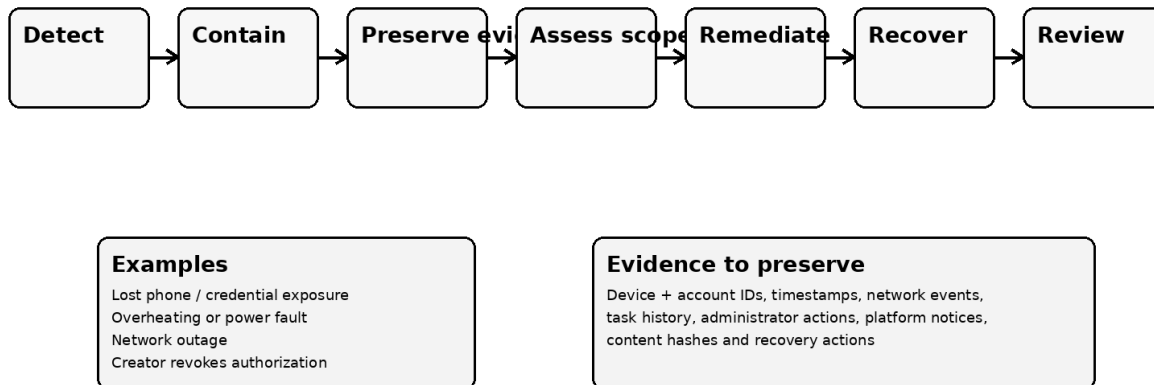


Figure 57-1. Incident-response sequence.

Seven stages

Detect -> contain -> preserve evidence -> assess scope -> remediate -> recover -> review. For a lost phone, containment may include MDM lock/wipe and secret revocation. For overheating, containment is physical power isolation and safe handling. For a content leak, containment includes access revocation and distribution/takedown actions consistent with law and platform processes.

Maintain an incident commander role so multiple employees do not issue conflicting changes.

- Incident ID and severity.
- Affected creator/account/device.
- Timeline.
- Actions/actors.
- Evidence.
- External notifications if required.
- Root cause and corrective action.

Testing

Run tabletop exercises for: lost device, compromised operator credential, database outage, storage exposure, creator revokes access, network failure during publishing, and rack thermal alarm.

58. Privacy and data minimization

A growth/fleet system should not become an uncontrolled surveillance database. Collect the minimum data needed to operate and secure the business.

Minimize

Device health does not require reading personal messages. Network health does not require storing full destination histories indefinitely. Analytics often need campaign/link IDs rather than raw IP addresses. Separate security logs from marketing data and apply different retention policies.

If employees operate creator accounts, document monitoring and privacy expectations in employment/contract policy and comply with applicable labor/privacy law.

- Purpose for each field.
- Retention period.
- Access role.
- Encryption requirement.
- Deletion/export process.

Data subject/creator exit

Be able to locate creator-specific records and assets for contractual deletion/return while preserving records the agency is legally required to retain. Avoid schemas that mingle multiple creators' sensitive content in untraceable folders.

59. Business continuity and creator exit

Continuity is not only infrastructure uptime; it includes a clean way for a creator to leave without the business or creator losing control of accounts.

Creator exit

Freeze new tasks, inventory pending posts, return/transfer authorized credentials or delegated roles according to contract/platform capabilities, revoke agency sessions, remove employee access, stop scheduled content, export agreed analytics/assets, and wipe creator material from reassigned devices.

Confirm recovery email/phone and MFA ownership before disabling agency access so the creator is not accidentally locked out.

- Stop queue.
- Reconcile scheduled/published content.
- Transfer/revoke access.
- Export/return agreed assets.
- Delete/wipe according to policy.
- Record completion.

Continuity

Document who can operate if the primary fleet admin is unavailable. Store network/MDM recovery procedures securely. At least two appropriately authorized people should be able to recover the system without sharing one administrator password.

PART VIII - SCALING AND IMPLEMENTATION

Scale in stages. A ten-device pilot should prove enrollment, physical design, network stability, queueing, attribution and incident recovery before the agency buys fifty more phones.

60. Scaling from 5 to 100+ devices

Scaling changes the bottleneck. At five devices, manual labeling and one AP may be adequate. At fifty, power zones, Wi-Fi capacity, management automation and staffing dominate. At one hundred, redundancy, change control and inventory discipline become mandatory.

Stage 1: 5-10

Goal: validate workflow. Use complete standard phones, one or two device models, managed Wi-Fi, a small controller and basic MDM/Android Enterprise. Measure actual power, heat, posting labor and conversion.

Do not optimize rack density yet. Optimize observability.

Stage 2: 25-50

Add segmented VLANs, multiple planned AP cells as needed, rack zones, UPS-backed network/control plane, automated enrollment, formal content approvals, task queue, metrics and scheduled maintenance. Separate backend/database from heavy media processing if necessary.

Stage 3: 100+

Use multiple physical racks/zones, standardized spares, automated inventory reconciliation, redundant network edge, tested disaster recovery, role-separated staff and canary change rings. Consider multiple sites only after one site is operationally mature.

Dimension	5-10	25-50	100+
Enrollment	Manual assisted	MDM/Enterprise standardized	Automated procurement/enrollment
Power	Single measured zone	Multiple protected zones	Rack/site capacity plan
Wi-Fi	1 planned AP may suffice	Capacity-driven multi-AP	RF design + monitoring
Controller	Single host	Separated roles as needed	HA services
Staffing	1-2 cross-functional	Defined roles	Shift + incident ownership

61. Cost model and BOM categories

Total cost is CAPEX plus recurring OPEX. A cheap phone can be expensive if it requires twice the labor or fails frequently.

CAPEX

Phones, rack/enclosure, chargers/PD supplies, cables, PDU, UPS, router/firewall, managed switch, APs, controller/workstations, environmental sensors, spare devices and installation materials.

Price bands vary strongly by country and used-device market. Obtain current local quotes rather than relying on a static web price in this manual.

- Devices.
- Rack/trays.
- Power distribution.
- Network.
- Controller/storage.
- Spares.
- Safety/environmental monitoring.

OPEX

Internet/cellular plans, MDM licenses, cloud hosting, object storage/egress, monitoring, domain/link infrastructure, employee labor, repairs/batteries, platform/developer costs where applicable, legal/compliance and backup storage.

Allocate shared infrastructure by campaign or creator using a transparent model. Labor is often the largest cost in a device-first workflow.

Cost class	Driver	Track as
Phone hardware	Device count/lifecycle	Monthly amortization
Power/network	Rack/site	Shared infrastructure
Software/MDM	Managed devices/users	Recurring license
Cloud/storage	GB, requests, compute	Usage
Connectivity	WAN/SIM plans	Monthly recurring
Labor	Tasks/hours	Direct operating cost

62. Pilot build - 10 device reference

The reference pilot is deliberately conservative. Its goal is to prove the system and gather measurements, not maximize density.

Hardware categories

Ten organization-owned phones from one or two supported model families; a ventilated non-compressing rack/tray; reputable multi-port charging or zoned chargers; labeled data cables for the subset needing USB admin; business firewall/router; managed PoE switch; one appropriately placed business AP; controller workstation; small UPS for network/controller; temperature sensor; spare phone and cables.

Do not purchase dedicated proxy infrastructure for the pilot unless there is a legitimate documented networking requirement. Direct business internet with proper segmentation is simpler to debug.

Software

MDM/Android Enterprise, inventory database, credential vault, basic coordinator/task board, object storage, tracking-link service, metrics/logging and a weekly report. TikTok or other official APIs can be integrated only after platform application authorization requirements are understood.

Pilot exit criteria

Every device can be enrolled/recovered, every account is authorized, every task is attributable, network/thermal behavior is stable, backup restore works, creator exit can be executed, and the business can calculate cost per successful campaign output.

Engineering checklist

- Run 72-hour full-load thermal/network test.
- Simulate one AP/router/controller restart.
- Wipe and re-enroll one device.
- Revoke one operator and verify access is gone.
- Publish/verify through every supported path.
- Restore database from backup.

63. 25-50 device expansion

Expansion is justified only after pilot measurements show stable operations and positive business value.

Physical expansion

Divide the rack into power/service zones, add airflow based on measured heat, add AP capacity based on measured airtime/utilization, and maintain at least 5-10% spare device capacity. Use port/slot maps and environmental alerting.

If a single controller still performs ADB, video transcoding, database and dashboard functions, split those roles before they become a single point of performance and failure.

Process expansion

Introduce formal change management: canary devices, scheduled OS/app updates, weekly access review for high-privilege roles, incident severity levels and capacity dashboards. Add a second trained administrator before the system becomes dependent on one person.

Software expansion

Move task execution to asynchronous queues, implement idempotency, add metrics/alerting, automate inventory reconciliation and make the creator/account authorization model explicit in every API request.

64. 30/60/90 day implementation roadmap

The roadmap assumes the agency starts with no mature device-management control plane. Adjust based on existing infrastructure.

Days 0-30 - prove the foundation

Inventory creators/accounts; define policy boundary; select 5-10 devices; deploy network segmentation; establish MDM/Android Enterprise; create rack/power/thermal pilot; deploy credential vault; implement creator/account/device/asset database; create manual task queue and tracking links; record baseline costs.

Deliverables: physical pilot, authorization register, network diagram, enrollment SOP, incident contact list, first weekly funnel report.

Days 31-60 - automate administration

Build coordinator API and companion/dashboard; add task state machine, approvals, object storage, health telemetry, metrics/alerts, canary updates and backup restore tests. Integrate one supported publishing API such as TikTok Content Posting API if business requirements and app authorization permit.

Deliverables: auditable task flow, fleet dashboard, monitoring, restore evidence, controlled API publishing path.

Days 61-90 - scale and govern

Expand to 25-50 only if KPIs justify it. Add power zones/AP capacity, formal roles, weekly access reviews, incident exercises, creator exit test, cost allocation, performance experiment process and quarterly platform-policy review.

Deliverables: scaled rack, runbooks, staff training, policy register, cost/conversion dashboard and go/no-go criteria for further scale.

65. Acceptance test plan

The installation is not complete because the phones turn on. Acceptance tests prove the physical, network, management, security and business workflows.

Physical/electrical

Measure AC load at idle and representative peak, verify no supply exceeds its rating, confirm rack temperatures in normal and degraded fan conditions, inspect cable strain and confirm emergency disconnect. Run a controlled power failure and verify UPS behavior for network/control systems.

Network/fleet

Verify DHCP/DNS/VLAN policy, Wi-Fi stability, tunnel/proxy behavior if configured, management reachability, enrollment, remote lock/wipe or equivalent supported actions, OS/app inventory and recovery after reboot.

Business/security

Create an approved asset, assign a task, publish via each supported path, capture post ID, verify tracked link, ingest analytics, produce weekly report, revoke an operator, remove a creator, restore a backup and execute one incident tabletop.

Engineering checklist

- All test evidence attached to acceptance record.
- Known exceptions have owner and due date.
- No production scaling before critical failures are closed.

APPENDIX A - Example IP/VLAN plan

Example only. Choose ranges that do not conflict with the organization's existing network or VPNs.

VLAN	Purpose	Example subnet	Allowed inbound	Primary egress
10	Management	10.20.10.0/24	Admin/VPN only	Direct corporate
20	Phone fleet A	10.20.20.0/24	Mgmt services only	Policy-defined
21	Phone fleet B	10.20.21.0/24	Mgmt services only	Policy-defined
30	Data/storage	10.20.30.0/24	API/worker only	Restricted
40	Quarantine	10.20.40.0/24	Mgmt only	Updates/support only
50	Guest	10.20.50.0/24	None	Direct isolated

Do not expose MDM connectors, ADB services, broker ports, database or storage administration directly to the public internet. Administrative access should traverse a secured management path.

APPENDIX B - Example firewall policy

Source	Destination	Service	Action	Reason
Mgmt VLAN	Fleet API	HTTPS	Allow	Admin/dashboard
Device VLAN	DNS resolvers	DNS/DoH/DoT as designed	Allow	Name resolution
Device VLAN	NTP/time	NTP/HTTPS time	Allow	Clock sync
Device VLAN	Internet	HTTPS + required platform traffic	Allow	Workload
Device VLAN	Mgmt VLAN	Any	Deny	Prevent lateral access
Device VLAN	Device VLAN	Any	Deny by default	Limit east-west
Quarantine	Internet	Updates/support subset	Allow limited	Recovery
Internet	Database/ADB/broker admin	Any	Deny	No direct exposure

The exact service list depends on Apple/Google management requirements, VPN/MDM vendor endpoints and platform applications. Use vendor-published network requirements and validate with staged tests.

APPENDIX C - Task JSON and API examples

These examples describe the internal coordinator. They do not contain commands for automating third-party social user interfaces.

```
POST /v1/tasks
{
  "creator_id": "cr_123",
  "account_id": "acct_tiktok_456",
  "asset_version_id": "assetv_789",
  "action": "PUBLISH_APPROVED_ASSET",
  "executor_type": "OFFICIAL_API_OR_HUMAN",
  "approval_ids": ["appr_001"],
  "campaign_id": "cmp_2026_08_a",
  "not_before": "2026-08-27T12:00:00Z",
  "deadline": "2026-08-27T18:00:00Z",
  "idempotency_key": "cmp...acct...asset..."
}
```

```
POST /v1/tasks/{id}/result
{
  "status": "VERIFIED",
  "executor_id": "operator_or_connector_id",
  "device_id": "dev_012",
  "platform_post_id": "platform-returned-id",
  "completed_at": "...",
  "evidence": {"type": "platform_post_reference"}
}
```

Recommended API safeguards

- OAuth/OIDC for employees; service credentials for connectors.
- Role and creator/account scope on every request.
- Idempotency keys for create/publish requests.
- Request ID and actor in every audit event.
- Schema validation and bounded text lengths.
- No plaintext platform passwords in request bodies.
- Rate limits for internal clients.

APPENDIX D - Example SQL data model

```
CREATE TABLE device (
  id UUID PRIMARY KEY, asset_tag TEXT UNIQUE NOT NULL,
  platform TEXT NOT NULL, model TEXT, serial_ref TEXT,
  management_state TEXT NOT NULL, rack_slot TEXT,
  last_heartbeat TIMESTAMPTZ, created_at TIMESTAMPTZ NOT NULL
);

CREATE TABLE platform_account (
  id UUID PRIMARY KEY, creator_id UUID NOT NULL, platform TEXT NOT NULL,
  handle TEXT, owner_type TEXT, authorization_ref TEXT NOT NULL,
  credential_ref TEXT, status TEXT NOT NULL
);

CREATE TABLE media_asset_version (
  id UUID PRIMARY KEY, creator_id UUID NOT NULL, source_asset_id UUID,
  sha256 TEXT NOT NULL, storage_uri TEXT NOT NULL,
  rights_ref TEXT NOT NULL, qc_status TEXT NOT NULL
);

CREATE TABLE task (
  id UUID PRIMARY KEY, creator_id UUID NOT NULL, account_id UUID NOT NULL,
  asset_version_id UUID, action TEXT NOT NULL, state TEXT NOT NULL,
  assigned_device_id UUID, assigned_operator_id UUID,
  idempotency_key TEXT UNIQUE, created_at TIMESTAMPTZ NOT NULL
);
```

Store sensitive serial/IMEI/recovery metadata with access controls or separate encrypted references as appropriate. The schema is illustrative and omits foreign keys/indexes for readability.

APPENDIX E - MQTT topic design

fleet/{device_id}/health	retained summary, device -> broker
fleet/{device_id}/events	transient events, device -> broker
fleet/{device_id}/commands	narrow approved internal commands
fleet/{device_id}/command-result	result acknowledgements
racks/{rack_id}/environment	temperature/power sensors
ops/incidents	security/operations event feed

Authorization model

- A device may publish only to its own health/events/result topics.
- A device may subscribe only to its own command topic.
- Rack sensors may publish only to their rack namespace.
- Operators use the API, not direct broker credentials.
- Broker admin interface is management-network only.
- TLS client authentication or strong per-device credentials.

APPENDIX F - Device commissioning checklist

Receiving

- ☐ Asset tag assigned
- ☐ Serial/model recorded
- ☐ Activation/carrier locks checked
- ☐ Battery/charging port inspected
- ☐ Factory reset performed

Enrollment

- ☐ Correct owner/enterprise
- ☐ MDM/Android Enterprise enrollment complete
- ☐ Supervision/fully-managed state verified
- ☐ Baseline policy applied
- ☐ Required apps installed

Network

- ☐ Correct SSID/VLAN
- ☐ DNS works
- ☐ VPN/proxy policy verified if applicable
- ☐ Management heartbeat received
- ☐ Public egress recorded for diagnostics

Security

- ☐ Passcode/lock policy
- ☐ Disk/device encryption active by platform default/management
- ☐ Debugging disabled unless required
- ☐ Credential vault assignment complete
- ☐ Lost/wipe procedure tested on cohort

Operations

- ☐ Rack slot assigned
- ☐ Cable/port mapped
- ☐ Tracking/companion app check-in
- ☐ Test task completed
- ☐ Status set to Ready

APPENDIX G - Weekly operations report template

Section	Required fields
Executive summary	What improved, what deteriorated, major risk/blocker, next decision
Fleet health	Online %, degraded/offline devices, thermal/power incidents, update status
Publishing	Tasks completed/failed, failure causes, approval latency, executor split
Account health	Restrictions/notices, appeals, account availability, policy changes
Funnel	Reach where available, clicks, conversion, revenue attribution confidence
Content	Top/bottom assets, experiments, QC/rejection rate
Security	Access changes, incidents, credential rotations, overdue reviews
Cost	Labor hours, connectivity/software, repair/amortization, estimated contribution
Next week	Priorities, experiments, maintenance, owner and deadline

APPENDIX H - Incident runbooks

Lost or stolen phone

1. Mark device Quarantined in coordinator.
2. Use MDM/Enterprise controls to lock/wipe as appropriate.
3. Revoke internal device token and any account sessions/credentials accessible from the device.
4. Preserve inventory/last-seen/network/task logs.
5. Notify creator/security owner if their account/content was exposed.
6. Replace from spare pool and reassign only after incident review.

Credential compromise

7. Disable/revoke compromised account or delegated session.
8. Rotate password/API token and MFA recovery material as required.
9. Search audit logs for access/use during exposure window.
10. Revoke employee/service sessions that inherited the secret.
11. Restore only after clean admin workstation and access path are verified.

Rack overheating

12. Stop new workload.
13. Inspect fan/airflow and ambient conditions.
14. Isolate abnormally hot/swollen devices; disconnect safely.
15. Reduce charging load or power down affected zone.
16. Do not return devices until temperatures and root cause are normal.

Platform restriction

17. Preserve notice and post/account IDs.
18. Pause similar queued tasks.
19. Classify policy/security/copyright/age-consent cause.
20. Use legitimate remediation/appeal path.
21. Update campaign rules; do not auto-create replacement identities.

Database outage

22. Pause writes/publishing that cannot be reconciled.
23. Verify storage/host health.
24. Fail over or restore according to runbook.
25. Reconcile queue vs database after recovery.
26. Document RPO/RTO achieved and any lost events.

APPENDIX I - Capacity formulas

Electrical planning

```
P_devices_avg = N_devices x measured_average_watts_per_device
P_total_avg   = P_devices_avg + P_network + P_controller + P_fans + P_storage
P_design      = P_total_expected x engineering_headroom_factor
```

Example only:

25 phones x 5 W measured average = 125 W device load

+ 120 W shared infrastructure = 245 W average

Design supplies/circuit/UPS from measured peak and required headroom, not this example.

Storage planning

```
Daily_media_GB = assets_per_day x average_asset_MB / 1024
Retention_GB   = Daily_media_GB x retention_days x version_multiplier
Backup_GB      = Retention_GB x backup_copy_factor
```

Add database, logs and temporary transcoding space separately.

Task throughput

```
Human_capacity_tasks_per_shift = operators x productive_minutes_per_shift / median_task_minutes
Required_concurrency_API       = peak_tasks_per_minute x p95_execution_minutes
```

Always apply platform/API rate limits and quality constraints before raw capacity targets.

Wi-Fi

Do not use a single “devices per AP” constant. Capacity depends on airtime, channel width, modulation/rate, interference, application traffic, AP radio design and environment. Measure channel utilization, retries and application latency under representative concurrent load.

APPENDIX J - Glossary

Term	Meaning
ADB	Android Debug Bridge; Google development/diagnostic bridge for authorized Android devices.
ACL	Access Control List; rule set permitting/denying network flows.
AP	Wireless access point.
APNs	Apple Push Notification service; used by MDM to signal devices that management work is available.
CGNAT	Carrier-Grade Network Address Translation; NAT performed in a carrier network at subscriber scale.
Control plane	Systems that manage policy, tasks, identity, inventory and administration.
Data plane	Actual media/application/network traffic produced by workload execution.
Dedicated device	Android Enterprise fully managed device locked to one or a small set of work purposes/apps.
DoH	DNS over HTTPS.
DoT	DNS over TLS.
Egress	Outbound path/address through which traffic leaves a network.
Idempotency	Property that allows the same logical request to be retried without creating duplicate effects.
MDM	Mobile Device Management.
MQTT	Lightweight publish/subscribe messaging protocol.
NAT	Network Address Translation.
PDU	Power Distribution Unit.
Proxy	Intermediary that makes/relays application connections on behalf of a client.
RBAC	Role-Based Access Control.
RPO	Recovery Point Objective; maximum tolerable data loss window.
RTO	Recovery Time Objective; target service restoration time.
Supervision	Apple organization-owned device management state providing additional controls.
TUN	Virtual network interface commonly used by VPN software.
VLAN	Virtual LAN; logical Layer-2 network segmentation.
VPN	Virtual Private Network; encrypted/logical tunnel to a network gateway.

APPENDIX K - Research references

Research cut: 26 August 2026. First-party sources are preferred. Secondary/archived sources are labeled. Platform policies can change; verify again before implementation.

ID	Source	URL / note	Quality
S1	Google Android Management API - Enroll and provision a device	https://developers.google.com/android/management/provision-device	First-party
S2	Google Android Management API - Policies reference	https://developers.google.com/android/management/reference/rest/v1/enterprises.policies	First-party
S3	Android Developers - Android Debug Bridge (adb)	https://developer.android.com/tools/adb	First-party
S4	Android Developers - VPN / VpnService	https://developer.android.com/develop/connectivity/vpn	First-party
S5	Apple Platform Deployment	https://support.apple.com/guide/deployment/welcome/web	First-party
S6	Apple Developer - Device Management	https://developer.apple.com/documentation/devicemanagement/	First-party
S7	Apple Developer - MDM commands and queries	https://developer.apple.com/documentation/devicemanagement/commands-and-queries	First-party
S8	Apple Platform Deployment - supervision	https://support.apple.com/guide/deployment/dep1d89f0bff/web	First-party
S9	AWS Device Farm Developer Guide	https://docs.aws.amazon.com/devicefarm/latest/developerguide/welcome.html	First-party; comparison/testing reference
S10	USB-IF - USB Charger / USB Power Delivery	https://www.usb.org/usb-charger-pd	First-party standards body
S11	Apple - Keeping iPhone/iPad within acceptable operating temperatures	https://support.apple.com/en-gb/118431	First-party
S12	Apple - iPhone battery and charging	https://support.apple.com/en-gb/105105	First-party
S13	Cisco - Wireless High Client Density Design Guide	https://www.cisco.com/c/en/us/td/docs/wireless/controller/technotes/8-7/b_wireless_high_client_density_design_guide.html	First-party vendor design guide
S14	Cloudflare - Forward/reverse proxy explanation	https://www.cloudflare.com/learning/cdn/glossary/reverse-proxy/	Vendor educational reference
S15	Cloudflare - A primer on proxies	https://blog.cloudflare.com/a-primer-on-proxies/	Vendor technical reference
S16	Cloudflare - DNS security	https://www.cloudflare.com/learning/dns/dns-security/	Vendor educational reference
S17	Cloudflare - DNS over TLS vs DNS over HTTPS	https://www.cloudflare.com/learning/dns/dns-over-tls/	Vendor educational reference
S18	APNIC - Carrier Grade NAT training material	https://training.apnic.net/wp-content/uploads/sites/2/2016/12/	Regional Internet registry training

		TIP602.pdf	
S19	WireGuard official site	https://www.wireguard.com/	First-party project
S20	MQTT official site	https://mqtt.org/	Standards/project reference
S21	Prometheus - Overview	https://prometheus.io/docs/introduction/overview/	First-party project
S22	TikTok for Developers - Content Posting API	https://developers.tiktok.com/doc/content-posting-api-get-started	First-party
S23	TikTok Community Guidelines - Integrity and Authenticity	https://www.tiktok.com/community-guidelines/en/integrity-authenticity/	First-party
S24	Reddit Rules	https://redditinc.com/policies/reddit-rules	First-party
S25	Reddit Data API Terms	https://redditinc.com/policies/data-api-terms	First-party
S26	Snap - Harmful False or Deceptive Practices	https://values.snap.com/policy/policy-community-guidelines/harmful-false-deceptive-practices	First-party
S27	X Help - Authenticity	https://help.x.com/en/rules-and-policies/authenticity	First-party
S28	UK Parliament written evidence quoting historical OnlyFans terms	https://committees.parliament.uk/writtenevidence/40780/default/	Official public record; historical
S29	Archived OnlyFans Terms snapshot	https://archive.ph/6mux7	Archive/secondary; verify current terms directly
S30	2026 secondary analysis of OnlyFans agency terms	https://whalefinders.com/blog/does-onlyfans-allow-agencies-tos-rules-2026	Secondary; unverified interpretation
S31	OSHA - Extension Cords and Power Strips Safety at Work (2026)	https://www.osha.gov/sites/default/files/publications/OSHA4495.pdf	US government safety guidance

Anecdotal industry material

During research, public OFM/phone-farm discussions were reviewed to understand terminology and claimed operating patterns. Self-described operators discuss old iPhone racks, multi-account posting, proxies, UI automation and other tactics. These reports are not independently verified, and some described tactics appear designed to bypass platform integrity controls. They are therefore not reproduced as implementation instructions in this guide. The architecture instead uses enterprise device-management concepts, approved APIs, human workflows and auditable networking.

Final engineering principle

Make the fleet observable, replaceable and accountable

A professional system should be able to answer, at any moment: Who authorized this account? Which approved asset was used? Which employee or connector executed the task? Which device/network path was involved? What result occurred? Can access be revoked immediately? Can the device be wiped and replaced? Can the business show whether the activity generated value? If those questions cannot be answered, adding more phones increases risk faster than capacity.

Source-use matrix

Topic	Primary sources
Android enrollment/policy/ADB/VPN	S1-S4
Apple enrollment/supervision/MDM	S5-S8
Device-farm comparison	S9
Power/thermal/safety	S10-S13, S31
Proxy/DNS/CGNAT/VPN	S14-S19
Messaging/monitoring	S20-S21
TikTok publishing/policy	S22-S23
Reddit/Snap/X rules	S24-S27
OnlyFans responsibility context	S28-S30 (historical/secondary caveat)